

Berry-Landau Fermi-liquid theory: transport in presence of quantum geometry

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Landau Fermi-liquid theory characterizes interacting metals through quasiparticles and their residual interactions. It is a challenge to incorporate non-trivial quantum geometry – as encoded by Berry phases and the quantum metric – as a fundamental ingredient. We formulate a Berry-Landau Fermi-liquid theory for spinless fermions within an isolated band crossing the Fermi surface and derive the Landau functional through the Nozières–Luttinger construction. The instantaneous response to a perturbation of the Bloch waves generates both an anomalous Berry-connection potential in the action and an interaction-induced quantum-geometric contribution to the quasiparticle current. The conserved physical charge current is then obtained via the electromagnetic Ward identity as a combination of this quantum-geometric current and the bare drift current. Therefore, the intrinsic anomalous Hall conductivity is fixed by the Berry-curvature integral of the dressed quasiparticle band while the Drude weight contains both conventional and quantum-geometric contributions. In the presence of Galilean symmetry, the Drude weight is protected against interaction renormalization. In the flat/narrow-band limit, transport is dominantly quantum-geometric and can be thermally enhanced. These results establish quasiparticle occupations, Landau interactions, and the quantum geometry carried by quasiparticles as the fundamental low-energy ingredients of a Berry-Landau Fermi liquid.

Introduction.— Landau’s Fermi-liquid (FL) theory describes an interacting metal through long-lived quasiparticles labeled by momentum, with the low-energy state specified by their occupations. In the presence of non-trivial quantum geometry (QG), this needs to be supplemented by information about the Berry phase and Hilbert-space geometry [1–8]. Far from passive, this geometry controls transport—the anomalous velocity, orbital magnetization, linear and nonlinear Hall responses, and positional shifts of the carriers [9–19]—and enters the interaction itself through the form factors of band projection, driving, e.g., flat-band superconductivity governed by the quantum metric [20–30]. Quantum-geometric transport is realized in topological semimetals [31–33], kagome and pyrochlore metals [34–36], and moiré materials [37–39], and QG is now measurable [40, 41]. The question thus arises of how QG is most naturally incorporated into FL theory.

The notion that it is the Fermi surface which determines transport properties extends to topological transport phenomena: the Berry connection on the Fermi surface determines the intrinsic anomalous Hall effect (AHE) [8, 9, 42–44]. Berry-curvature effects enter kinetic theory at the single-particle level [45, 46] and have been combined with interaction-dressed vertices in FL transport [47–51]. In these approaches, QG is imported from the underlying multiband structure or postulated in a phenomenological action, rather than constructed from just the Fermi surface itself. Recently, we proposed a generalized Peierls substitution [25]. There, we noted that the current density operator acquires a QG contribution in the presence of interactions. In this work, we show how to incorporate into FL theory such QG effects, alongside the standard ingredients of quasiparticle occupations and Landau parameters.

Towards a microscopic formulation of FL theory, the Nozières–Luttinger construction establishes that the Landau functions encode the residual forward scattering between quasiparticles [52, 53] through the particle-hole Bethe-Salpeter equation [54–57]. Adapting this construction, we develop a Berry-Landau-FL (BL-FL) theory by projecting an interacting multiorbital lattice model onto an isolated band, and extract physical consequences such as the resulting transport properties. Finally, we verify these predictions using Hartree-Fock calculations for an isolated Chern band of the Wilson-Dirac model [58–60].

System setup.— To treat quantum-geometric effects in an interacting metal, we start with a generic N_{orb} -orbital lattice Hamiltonian $H_0 = \sum_{\mathbf{k}} H_0(\mathbf{k})$ with

$$H_0(\mathbf{k}) = \sum_{\alpha, \beta=1}^{N_{\text{orb}}} t_{\alpha\beta}(\mathbf{k}) c_{\mathbf{k}\alpha}^\dagger c_{\mathbf{k}\beta}, \quad (1)$$

where $c_{\mathbf{k}\alpha}$ annihilates a fermion with crystal momentum $\mathbf{k}$ and orbital index α . Among the N_{orb} bands, we assume that H_0 hosts an isolated (active) band crossing the Fermi energy, separated from all remote bands by an energy gap Δ . Throughout this work, we take the isolated-band limit, $\Delta \rightarrow \infty$, at fixed active-band width, interaction strength, and temperature. We allow for the active band to be topologically nontrivial. We thus project onto the Hilbert subspace $\mathbb{H}^{\text{sub}}$ of the active band. A general short-ranged interaction is considered for the development of the theory.

Anomalous potential from band projection without interactions.— Consider first the noninteracting problem. The Hilbert subspace $\mathbb{H}^{\text{sub}}$ is spanned by the active band at each crystal momentum $\mathbf{k}$, $H_0(\mathbf{k})|u_{\mathbf{k}}\rangle = \epsilon_{\mathbf{k}}^0|u_{\mathbf{k}}\rangle$. Then the projection onto $\mathbb{H}^{\text{sub}}$ can be imposed via a

Grassmann-valued Lagrange multiplier $\lambda_{\mathbf{k}\alpha}$ [25],

$$S_\lambda = \int dt \sum_{\mathbf{k}\alpha} \left[\lambda_{\mathbf{k}\alpha}^\dagger (c_{\mathbf{k}\alpha} - u_{\mathbf{k}\alpha} f_{\mathbf{k}}) + \text{h.c.} \right], \quad (2)$$

where the operator $f_{\mathbf{k}}$ annihilates a fermion at crystal momentum $\mathbf{k}$ of the active band. We have the partition function $Z = \int Dc Dc^\dagger Df Df^\dagger D\lambda D\lambda^\dagger e^{i(S_0 + S_\lambda)}$ with the action $S_0 = \int dt \sum_{\mathbf{k}} [\sum_{\alpha} c_{\mathbf{k}\alpha}^\dagger i\partial_t c_{\mathbf{k}\alpha} - H_0(\mathbf{k})]$. In the absence of external perturbations, integrating out λ yields the free theory, $S[f] = \int dt \sum_{\mathbf{k}} f_{\mathbf{k}}^\dagger (i\partial_t - \epsilon_{\mathbf{k}}^0 + \mu) f_{\mathbf{k}}$.

QG manifests itself through the response to external perturbations. Taking a uniform electric field as an example, the Bloch waves become explicitly time dependent, and this time dependence encodes the Berry connection. We choose the gauge $\mathbf{A}(t) = -\mathbf{E}t$, such that $H_0(\mathbf{k}, t) = H_0(\mathbf{k} + e\mathbf{A}(t))$. In the weak-field limit, the instantaneous Bloch wave follows the Houston trajectory [61, 62], $|u_{\mathbf{k}}(t)\rangle = |u_{\mathbf{k}(t)}\rangle$ with $\dot{\mathbf{k}} = -e\mathbf{E}$. Accordingly, the projection in S_λ is imposed in the moving Bloch frame, $c_{\mathbf{k}\alpha} = u_{\mathbf{k}(t)\alpha} f_{\mathbf{k}}$. After projection, the time-derivative term produces an additional quantum-geometric contribution,

$$\sum_{\alpha} c_{\mathbf{k}\alpha}^\dagger i\partial_t c_{\mathbf{k}\alpha} \rightarrow f_{\mathbf{k}}^\dagger i\partial_t f_{\mathbf{k}} + e\mathbf{E} \cdot \mathbf{a}(\mathbf{k}) f_{\mathbf{k}}^\dagger f_{\mathbf{k}}, \quad (3)$$

with the Berry connection $\mathbf{a}(\mathbf{k}) = -i\langle u_{\mathbf{k}} | \nabla_{\mathbf{k}} | u_{\mathbf{k}} \rangle$. The nonzero value of $\dot{\mathbf{k}}$ is crucial for the appearance of the second term. For an electric field, it gives the Berry-connection contribution to the Wannier–Stark energy shift [63]. We refer to it as the anomalous potential because it generates the topological response, as we show later. The appearance of the anomalous potential is essential for gauge invariance due to the emergent gauge structure of the band projection [64].

The anomalous potential generates the intrinsic AHE,

$$\begin{aligned} \sigma_{xy} &= \frac{e^2}{\hbar} \int \frac{d^2\mathbf{k}}{(2\pi)^2} [-\partial_{\epsilon} n_F(\xi_{\mathbf{k}})] [v_{\mathbf{k}x}^0 a_y(\mathbf{k}) - v_{\mathbf{k}y}^0 a_x(\mathbf{k})] \\ &= \frac{e^2}{\hbar} \int \frac{d^2\mathbf{k}}{(2\pi)^2} n_F(\xi_{\mathbf{k}}) \Omega_{xy}(\mathbf{k}), \end{aligned} \quad (4)$$

where $\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}}^0 - \mu$ and $\Omega_{xy} = \partial_{k_x} a_y - \partial_{k_y} a_x$. The first line makes explicit the Fermi-surface character of the metallic AHE, consistent with Haldane’s formulation [8]. To see its origin, for a spatially uniform vector potential with $\mathbf{E}(\omega) = i\omega\mathbf{A}(\omega)$, integrating out f gives $S_{\text{eff}}[A] = -i \text{Tr} \ln(G_{f0}^{-1} - V[A])$, where $G_{f0}^{-1} = \omega - \epsilon_{\mathbf{k}}^0 + i0^+$, and $V[A] = e\mathbf{A} \cdot \mathbf{v}^0 - i\omega\mathbf{A} \cdot \mathbf{a}$. This defines the band and Berry-connection current operators, $\hat{\mathbf{J}}^{(v)} = e \sum_{\mathbf{k}} \mathbf{v}_{\mathbf{k}}^0 f_{\mathbf{k}}^\dagger f_{\mathbf{k}}$, and $\hat{\mathbf{J}}^{(B)} = -ie\omega \sum_{\mathbf{k}} \mathbf{a}(\mathbf{k}) f_{\mathbf{k}}^\dagger f_{\mathbf{k}}$. The intrinsic AHE arises from the current-current correlation between $\hat{\mathbf{J}}^{(v)}$ and $\hat{\mathbf{J}}^{(B)}$ in the Kubo response. More generally, any slow perturbation exerting a generalized force $\dot{\mathbf{k}}$ —a magnetic field, a smooth potential, a temperature gradient, strain,

or a frame rotation—generates the anomalous potential $\dot{\mathbf{k}} \cdot \mathbf{a}(\mathbf{k}) f_{\mathbf{k}}^\dagger f_{\mathbf{k}}$. The corresponding current operators and representative responses are summarized in Table I in the End Matter.

Adding interactions and the QG current to Landau theory.— Interactions play two roles: they modify the band structure and they generate an interaction-induced QG current [25]. The former determines the active Hilbert subspace. To capture this, we promote the Bloch waves $u_{\mathbf{k}\alpha}$ to variational parameters in Eq. (2). Stationarity identifies the retained state with the on-shell quasiparticle state,

$$G^{-1}(\omega, \mathbf{k})|u_{\mathbf{k}}\rangle = 0, \quad (5)$$

at the quasiparticle pole of the full Green function in the full Hilbert space. At $T = 0$, this construction is unambiguous on the Fermi surface, where the quasiparticle decay rate vanishes and the Hermitian matrix $G^{-1}(0, \mathbf{k}_F)$ has $|u_{\mathbf{k}_F}\rangle$ as its zero mode [8, 65, 66]. In the $\Delta \rightarrow \infty$ limit, the hybridization with remote bands is eliminated and the low-energy dynamics thus remains confined to this projected subspace.

Regarding the latter, interactions projected onto the subspace acquire form factors, whose dependence on an added perturbation as described above generates a contribution to the current operator. Before many-body renormalization, the projected interaction in the forward-scattering channel takes the form $H_{\text{int}} \rightarrow V^{-1} \sum_{\mathbf{k}\mathbf{k}'} f_{\mathbf{k}\mathbf{k}'}^0 n_{\mathbf{k}} n_{\mathbf{k}'}$, with $n_{\mathbf{k}} = f_{\mathbf{k}}^\dagger f_{\mathbf{k}}$. As the bare forward-scattering function $f_{\mathbf{k}\mathbf{k}'}^0$ contains the form factor, its response to a uniform vector potential gives the bare QG current operator $\hat{J}_i^{(\text{QG},0)} = \frac{e}{2V} \sum_{\mathbf{k}\mathbf{k}'} (\nabla_{\mathbf{k}} + \nabla_{\mathbf{k}'})_i f_{\mathbf{k}\mathbf{k}'}^0 n_{\mathbf{k}} n_{\mathbf{k}'}$. Many-body processes then renormalize the bare forward-scattering function $f_{\mathbf{k}\mathbf{k}'}^0$, which is the tree-level contribution to the amputated connected four-point vertex Γ , into the Landau function

$$f_{\mathbf{k}\mathbf{k}'} = Z_{\mathbf{k}} Z_{\mathbf{k}'} \lim_{\omega \rightarrow 0} \lim_{\mathbf{q} \rightarrow 0} \Gamma(\mathbf{k}, \mathbf{k}'; i\omega, \mathbf{q}), \quad (6)$$

where the dynamic limit is defined by taking the $\mathbf{q} \rightarrow 0$ limit before $\omega \rightarrow 0$. Correspondingly, this renormalized interaction generates the interaction correction to the bare QG current operator $\hat{J}^{(\text{QG},0)}$, yielding the renormalized QG current operator in the dynamic limit,

$$\hat{\mathbf{J}}^{(\text{QG},\omega)} = \frac{e}{2V} \sum_{\mathbf{k}\mathbf{k}'} (\nabla_{\mathbf{k}} + \nabla_{\mathbf{k}'}) f_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}} n_{\mathbf{k}'}. \quad (7)$$

This provides the foundation to construct the energy functional in the BL-FL theory.

Energy functional.— These results assemble into the BL-FL functional, $\delta E_{\text{BL-FL}}[n, \mathbf{A}] = \delta E_L[n] + \delta E_{\text{ext}}[n, \mathbf{A}]$. In the absence of explicit coupling to external perturbations, the first part retains the familiar Landau form,

$$\delta E_L[n] = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}}^{\text{qp}} \delta n_{\mathbf{k}} + \frac{1}{2V} \sum_{\mathbf{k}\mathbf{k}'} f_{\mathbf{k}\mathbf{k}'} \delta n_{\mathbf{k}} \delta n_{\mathbf{k}'}, \quad (8)$$

where $\delta n_{\mathbf{k}} = n_{\mathbf{k}} - n_{\mathbf{k}}^0$ and $\mathbf{v}_{\mathbf{k}}^{\text{qp}} = \nabla_{\mathbf{k}} \epsilon_{\mathbf{k}}^{\text{qp}}$. Here, a perturbation enters via a change in the occupation numbers.

The second term, upon coupling to external perturbations, is obtained by expanding around the equilibrium distribution (and dropping field-dependent constants):

$$\delta E_{\text{ext}}[n, \mathbf{A}] = \sum_{\mathbf{k}} \left[\mathbf{A} \cdot \mathcal{J}_L(\mathbf{k}) + \dot{\mathbf{k}} \cdot \mathbf{a}(\mathbf{k}) \right] \delta n_{\mathbf{k}} + \frac{1}{2V} \sum_{\mathbf{k}\mathbf{k}'} \mathbf{A} \cdot \mathbf{\Lambda}_{\mathbf{k}\mathbf{k}'} \delta n_{\mathbf{k}} \delta n_{\mathbf{k}'}, \quad (9)$$

with the quasiparticle drift current $e\mathbf{v}_{\mathbf{k}}^{\text{qp}}$, the longitudinal current $\mathcal{J}_L(\mathbf{k}) \equiv e \left[\mathbf{v}_{\mathbf{k}}^{\text{qp}} + \frac{1}{V} \sum_{\mathbf{k}'} n_{\mathbf{k}'}^0 \nabla_{\mathbf{k}'} f_{\mathbf{k}\mathbf{k}'} \right]$, and the QG kernel $\mathbf{\Lambda}_{\mathbf{k}\mathbf{k}'} \equiv e (\nabla_{\mathbf{k}} + \nabla_{\mathbf{k}'}) f_{\mathbf{k}\mathbf{k}'}$. Eq. (9) displays the three quantum-geometric structures of the BL-FL. The first is the dressed longitudinal current, including Landau backflow; the second is the anomalous Berry potential associated with the time dependence of the Bloch waves, with $\dot{\mathbf{k}}$ determined by the semiclassical force of the external perturbation; and the last is a bilinear QG coupling between occupation fluctuations. For an electromagnetic field, $\dot{\mathbf{k}} = -e(\mathbf{E} + \mathbf{v}_{\mathbf{k}}^{\text{qp}} \times \mathbf{B})$. The connection to the QG current operator in Eq. (7) is transparent. The coupling term between the gauge field and $\hat{\mathbf{J}}^{(\text{QG}, \omega)}$ is mapped as $\mathbf{A} \cdot \hat{\mathbf{J}}^{(\text{QG}, \omega)} \rightarrow \frac{1}{2V} \sum_{\mathbf{k}\mathbf{k}'} \mathbf{A} \cdot \mathbf{\Lambda}_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}} n_{\mathbf{k}'}$. Expanding $n_{\mathbf{k}} = n_{\mathbf{k}}^0 + \delta n_{\mathbf{k}}$, the term linear in $\delta n_{\mathbf{k}}$ gives the interaction-induced single-quasiparticle current, $\gamma^{(\text{int}), \omega}(\mathbf{k}) \equiv \frac{1}{V} \sum_{\mathbf{k}'} \mathbf{\Lambda}_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}'}^0$, so that $\mathcal{J}_L(\mathbf{k}) = e\mathbf{v}_{\mathbf{k}}^0 + \gamma^{(\text{int}), \omega}(\mathbf{k})$. The two derivatives in $\mathbf{\Lambda}_{\mathbf{k}\mathbf{k}'} = e(\nabla_{\mathbf{k}} + \nabla_{\mathbf{k}'}) f_{\mathbf{k}\mathbf{k}'}$ have the familiar Landau interpretation: the $\nabla_{\mathbf{k}}$ term gives the quasiparticle velocity shift $e(\mathbf{v}_{\mathbf{k}}^{\text{qp}} - \mathbf{v}^0)$, while the $\nabla_{\mathbf{k}'}$ term takes the Landau-backflow form after integration by parts over the Brillouin zone. The term quadratic in $\delta n_{\mathbf{k}} \delta n_{\mathbf{k}'}$ is the two-quasiparticle part of the QG current and contributes to nonlinear response and current fluctuations.

Charge conservation implies a Ward identity which fixes the longitudinal current $\mathcal{J}_L$ in relation to the bare current $e\mathbf{v}_{\mathbf{k}}^0$,

$$\mathcal{J}_L(\mathbf{k}) = e\mathbf{v}_{\mathbf{k}}^0 + \gamma^{(\text{int}), \omega}(\mathbf{k}). \quad (10)$$

Thus, $\mathcal{J}_L$ is the conserved physical current carried by a quasiparticle, including the interaction-induced velocity renormalization and Landau backflow. The Berry-connection current is transverse and is not constrained by this longitudinal identity. When Galilean invariance emerges in the long-wavelength limit, the Landau function depends only on the momentum transfer, $f_{\mathbf{k}\mathbf{k}'} = f(\mathbf{k} - \mathbf{k}')$, and thus $(\nabla_{\mathbf{k}} + \nabla_{\mathbf{k}'}) f_{\mathbf{k}\mathbf{k}'} = 0$. Therefore, the QG contribution vanishes and the physical current reduces to $\mathcal{J}_L = e\mathbf{v}^0$, recovering the conventional Landau relation.

The uniform collisionless response then follows from the free energy $F[n, \mathbf{A}] = \delta E_{\text{BL-FL}} - TS[n]$ with the entropy $S[n] = -\sum_{\mathbf{k}} [n_{\mathbf{k}} \ln n_{\mathbf{k}} + (1 - n_{\mathbf{k}}) \ln(1 - n_{\mathbf{k}})]$. Con-

cretely, we can obtain $F[n, \mathbf{A}]$ by substituting the linear-order quasiparticle energy variation $\delta \epsilon_{\mathbf{k}}^{\text{qp}} = \mathbf{A} \cdot \mathcal{J}_L(\mathbf{k}) + \dot{\mathbf{k}} \cdot \mathbf{a}(\mathbf{k}) + \frac{1}{V} \sum_{\mathbf{k}'} f_{\mathbf{k}\mathbf{k}'} \delta n_{\mathbf{k}'}$, with $\delta n_{\mathbf{k}} = \frac{\partial n_{\mathbf{k}}^0}{\partial \epsilon_{\mathbf{k}}^{\text{qp}}} \delta \epsilon_{\mathbf{k}}^{\text{qp}}$. Thus the longitudinal response is governed by $\mathcal{J}_L$, whereas the anomalous potential supplies the transverse quantum-geometric response. In particular, the Drude weight and intrinsic AHE follow, respectively, from the longitudinal $\mathbf{A}$ - $\mathbf{A}$ and antisymmetric $\mathbf{A}$ - $\mathbf{E}$ responses obtained from the free energy $F[n, \mathbf{A}]$.

AHE and transverse Wiedemann-Franz law.— The electric field produces the band-projected action term $e\mathbf{E} \cdot \mathbf{a}$, which is the Berry-connection contribution to the Wannier-Stark energy shift. Then differentiating the free energy $F[n, \mathbf{A}]$ with respect to $\mathbf{A}$ gives the physical current. Thus, we obtain the AHE response $\sigma_{xy}^{\text{AHE}} = \frac{e}{V} \sum_{\mathbf{k}\mathbf{k}'} \{ \mathcal{J}_{L,x}(\mathbf{k}) [(1 + \chi f)^{-1} \chi]_{\mathbf{k}\mathbf{k}'} a_y(\mathbf{k}') - (x \leftrightarrow y) \}$ where χ denotes the quasiparticle susceptibility with $\chi_{\mathbf{k}\mathbf{k}'} \equiv -\frac{\partial n_F}{\partial \epsilon_{\mathbf{k}}^{\text{qp}}} \delta_{\mathbf{k}\mathbf{k}'}$ and f denotes the Landau function $f_{\mathbf{k}\mathbf{k}'}$. The Ward identity in Eq. (10) then gives the AHE as a Fermi-surface integral

$$\sigma_{xy} = \frac{e^2}{V} \sum_{\mathbf{k}} \chi_{\mathbf{k}} \left[v_{\mathbf{k}x}^{\text{qp}} a_y(\mathbf{k}) - v_{\mathbf{k}y}^{\text{qp}} a_x(\mathbf{k}) \right], \quad (11)$$

with $\chi_{\mathbf{k}} = -\frac{\partial n_F}{\partial \epsilon_{\mathbf{k}}^{\text{qp}}}$. Thus, the Landau-backflow dressed current $\mathcal{J}_L(\mathbf{k}) = \sum_{\mathbf{k}'} (1 + \chi f)_{\mathbf{k}\mathbf{k}'} \mathbf{v}_{\mathbf{k}'}^{\text{qp}}$ cancels exactly against the Landau-backflow dressed Berry connection $\mathcal{A}(\mathbf{k}) = \sum_{\mathbf{k}'} [(1 + f\chi)^{-1}]_{\mathbf{k}\mathbf{k}'} \mathbf{a}(\mathbf{k}')$. Using $\chi_{\mathbf{k}} v_{\mathbf{k}i}^{\text{qp}} = -\partial_{k_i} n_F(\epsilon_{\mathbf{k}}^{\text{qp}} - \mu)$ and integrating by parts over the Brillouin zone yields

$$\sigma_{xy} = \sigma_{xy}^{\text{qp}} = \frac{e^2}{\hbar} \int_{\mathbf{k}} n_F(\epsilon_{\mathbf{k}}^{\text{qp}} - \mu) \Omega_{xy}^{\text{qp}}(\mathbf{k}). \quad (12)$$

The AHE is therefore the Berry-curvature integral of the quasiparticle band, with no additional Landau-backflow factor.

The transverse thermal Hall effect follows from introducing Luttinger's gravitational sources [67]. We introduce a gravitational vector potential $\mathbf{A}^g$ (conjugate to the heat current) which encodes the gravitoelectric field $\mathbf{E}^g = -\nabla T/T$. This gives the anomalous potential $-(\epsilon_{\mathbf{k}}^{\text{qp}} - \mu) \mathbf{E}^g \cdot \mathbf{a}(\mathbf{k})$, where the transported energy plays the role of the charge. Then, the longitudinal heat current is locked to the charge current, $\mathcal{J}_L^Q(\mathbf{k}) = \sum_{\mathbf{k}'} [(1 + f\chi)_{\mathbf{k}\mathbf{k}'} (\epsilon_{\mathbf{k}'}^{\text{qp}} - \mu) \mathbf{v}_{\mathbf{k}'}^{\text{qp}}]$. The intrinsic thermal Hall conductivity κ_{xy} is in turn obtained from the transverse heat-current response to the gravitoelectric field, i.e., the antisymmetric $\mathbf{A}^g$ - $\mathbf{E}^g$ response obtained from the free energy [68]

$$\kappa_{xy} = \frac{k_B^2 T}{\hbar} \int_{\mathbf{k}} \Omega_{xy}^{\text{qp}}(\mathbf{k}) I_2 \left(\frac{\epsilon_{\mathbf{k}}^{\text{qp}} - \mu}{T} \right), \quad (13)$$

where $I_2(a) = \int_a^\infty dy y^2 (-\frac{\partial n_F}{\partial y})$. Since $I_2(-\infty) = \pi^2/3$, comparison with Eq. (12) gives the transverse

Wiedemann-Franz (WF) law $\frac{\kappa_{xy}}{T\sigma_{xy}} \rightarrow L_0 = \frac{\pi^2 k_B^2}{3e^2}$ as $T \rightarrow 0$ [69, 70].

Drude weight.— On an isotropic Fermi surface the physical longitudinal current of Eq. (10) reduces to $\mathcal{J}_{L,i}(\mathbf{k}) = ev_{\mathbf{k}i}^{\text{qp}}(1 + F_1/d)$, with F_1 the $\ell = 1$ Landau parameter. With the quasiparticle current $e\mathbf{v}^{\text{qp}}$, we obtain the Landau form $D_{xx} = (1 + \frac{F_1}{d}) D_{xx}^{\text{qp}}$, where $D_{xx}^{\text{qp}} = \pi e^2 \int_{\mathbf{k}} \chi_{\mathbf{k}} [v_{\mathbf{k}}^{\text{qp}}]^2$ is the Drude weight associated with the quasiparticle dispersion. The total Drude weight can be decomposed into conventional and QG contributions,

$$D_{xx} = D_{xx}^{\text{con}} + D_{xx}^{\text{QG}}, \quad (14)$$

with $D_{xx}^{\text{con}} = \frac{v_F^0}{v_F^{\text{qp}}} D_{xx}^{\text{qp}}$, and $D_{xx}^{\text{QG}} = \frac{F_1^{\text{QG}}}{d} D_{xx}^{\text{qp}}$. Here v_F^0 and v_F^{qp} are the bare and quasiparticle Fermi velocities, respectively. We define the QG Landau parameter $\frac{F_1^{\text{QG}}}{d} \equiv \frac{1}{v_F^{\text{qp}}} \frac{1}{V} \sum_{\mathbf{k}'} \hat{\mathbf{k}} \cdot \Lambda_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}'}^0|_{k_F}$, which is the $\ell = 1$ component on the Fermi surface. Charge conservation then fixes the total through the sum rule $1 + \frac{F_1}{d} = \frac{v_F^0}{v_F^{\text{qp}}} + \frac{F_1^{\text{QG}}}{d}$. In a conventional FL whose Landau function depends only on the momentum transfer, the Ward identity fixes the physical current $\mathcal{J}_L(\mathbf{k}) = e\mathbf{v}_{\mathbf{k}}^0$. In a BL-FL, this relation can be broken: the quantum-geometric current in Eq. (7) exists already at $T = 0$, and is further modified by thermal redistribution. In a flat band with trivial QG ($v_F^0 = 0$), interactions can generate a finite quasiparticle velocity, but the quasiparticle current $e\mathbf{v}_{\mathbf{k}}^{\text{qp}}$ is exactly cancelled by Landau backflow, yielding $D_{xx}^{\text{cov}} = 0$. By contrast, a flat band with QG can exhibit transport of entirely quantum-geometric origin. A further feature of the BL-FL theory is a possible enhancement of transport with increasing temperature, as more states around the Fermi energy can contribute to the interaction-induced quasiparticle current, $\gamma_i^{(\text{int})\omega}(\mathbf{k}) = \frac{1}{V} \sum_{\mathbf{k}'} \Lambda_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}'}^0$, which in turn enters the physical current $\mathcal{J}_L$ in Eq. (10). We demonstrate this effect below for flat and narrow bands.

Numerical results.— We test the BL-FL predictions using self-consistent finite-temperature Hartree-Fock calculations for an isolated $|C| = 1$ Wilson-Dirac band [58–60] with screened density-density interactions of strength g [68]. We can independently control the bare bandwidth and the separation from the remote band while keeping the quantum geometry fixed. The isolated-band limit is realized by $\Delta \gg g, T, W_{\text{HF}}$, where W_{HF} is the bandwidth of the self-consistent HF band. Within HF, the quantum-geometric current in Eq. (7) is recognized by identifying the projected exchange kernel $\Sigma_{\mathbf{k}\mathbf{k}'}^{\text{HF}}$ with the Landau function $f_{\mathbf{k}\mathbf{k}'}$. The static HF self-energy defines the interacting Bloch basis, $h_{\text{HF}}(\mathbf{k})|u_{\mathbf{k}}^{\text{HF}}\rangle = \epsilon_{\mathbf{k}}^{\text{HF}}|u_{\mathbf{k}}^{\text{HF}}\rangle$, so that the band projection becomes $c_{\mathbf{k}\alpha} \rightarrow u_{\mathbf{k}\alpha}^{\text{HF}} f_{\mathbf{k}}$. The projected exchange kernel then generates the interaction-induced current $\gamma_i^{(\text{int}),\text{HF}}(\mathbf{k}) = \frac{e}{V} \sum_{\mathbf{k}'} n_{\mathbf{k}'}^0 (\nabla_{\mathbf{k}} + \nabla_{\mathbf{k}'})_i \Sigma_{\mathbf{k}\mathbf{k}'}^{\text{HF}}$, which is the HF realization of the quantum-geometric current in Eq. (7).

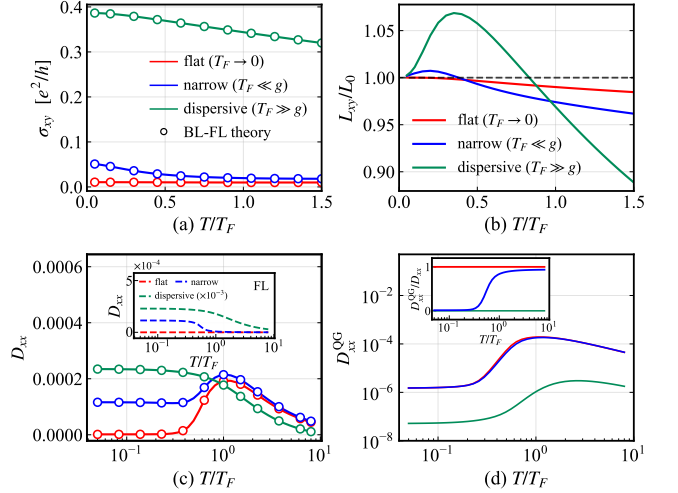


FIG. 1. Transport for three regimes in the Wilson-Dirac model: flat ($T_F \rightarrow 0$), narrow ($g/T_F = 91$), and dispersive ($g/T_F = 0.041$) from self-consistent finite-temperature HF. (a) AHE: full self-consistent HF for the response from twisting boundary conditions (solid) versus the Berry-curvature integral of the HF quasiparticle band (circles); agreement to $3 \times 10^{-6} e^2/h$. (b) Transverse Lorenz ratio L_{xy}/L_0 with the filled remote band restored; dashed line: L_0 . (c) Drude weight D_{xx} : Kohn stiffness from the twisted-boundary HF energy (solid) versus the prediction from BL-FL $D_{xx}^{\text{qp}}[1 + F_1(T)/2]$ (circles). Inset: conventional-FL controls; the flat-band control vanishes identically. (d) QG contribution D_{xx}^{QG} in Eq. (14); inset: the fraction $D_{xx}^{\text{QG}}/D_{xx}$. It equals one identically in the flat band, crosses over near $T \sim T_F$ in the narrow band, and is negligible in the dispersive band. Temperatures are normalized by the self-consistent Fermi scale T_F of each branch.

We study three regimes: flat ($T_F \rightarrow 0$), narrow ($g/T_F = 91$), and dispersive ($g/T_F \simeq 0.04$). Each is compared with a conventional-FL control having the same dispersion, gap, filling, and interaction, but with the Bloch-overlap factor $\langle u_{\mathbf{k}} | u_{\mathbf{k}'} \rangle$ in the projected interaction kernel replaced by unity.

Fig. 1(a) confirms the AHE in Eq. (12) within BL-FL theory: the self-consistent HF boundary-twist response, reconverged at each twist [see End Matter], agrees with the Berry-curvature integral of the thermal quasiparticle band to $3 \times 10^{-6} e^2/h$ for all regimes and temperatures. Panel (b) confirms the transverse WF law with the filled remote Chern band restored: all three ratios approach L_0 at zero temperature, while their finite-temperature behavior depends strongly on the band dispersion. The flat-band case remains closest to L_0 over a wide temperature range, decreasing only weakly to about $0.985L_0$ at $T = 1.5T_F$.

Panels (c,d) test the predictions on the Drude weight. We obtain the Drude weight from the Kohn stiffness, i.e., the curvature of the free energy with respect to a boundary twist. It agrees well with the relation from

the BL-FL theory at all plotted temperatures. For the flat band with trivial QG, the Drude weight remains identically zero in conventional FL theory, whereas the flat and narrow BL-FLs exhibit a finite thermal enhancement of the Drude weight, which peaks near $T \sim T_F$. This enhancement originates from the quantum-geometric current in Eq. (7). At low temperature, the occupied states form a small pocket around a band minimum, where both the quasiparticle velocity and the interaction-induced QG current are small. As temperature increases, thermal broadening allows more states to contribute to the QG current, enhancing the Drude weight. When $T > T_F$, the occupation becomes nearly uniform $n_{\mathbf{k}} \rightarrow \nu$, and the Drude weight crosses over to $D_{xx} \rightarrow \pi e^2 \nu (1 - \nu) \langle v_x^{\text{qp}} \mathcal{J}_{L,x} / e \rangle_{\text{BZ}} / k_B T$, which is consistent with the numerics. Panel (d) isolates the QG contribution to the Drude weight: the fraction $D_{xx}^{\text{QG}} / D_{xx}$ is identically 1 in the flat-band limit, crosses over from 0.013 to 0.94 near $T \sim T_F$ in the narrow band, and stays below 10^{-4} in the dispersive band.

Discussion.— We characterize two transport signatures in BL-FL theory: an intrinsic AHE determined by the quasiparticle Berry curvature and a QG contribution to the Drude weight. These responses highlight different aspects of quantum geometry: the AHE is controlled by the Berry curvature, whereas the longitudinal response reflects the interaction-induced QG current. The effects are especially pronounced in narrow bands, where interactions dominate over the band dispersion, allowing the QG current to become a leading contribution to transport. Narrow-band moiré systems therefore provide a natural setting in which these effects may be observable.

The framework also suggests several extensions. Spin or band degeneracy promotes the quantum geometry to a non-Abelian structure and can modify spin- and orbital-dependent Landau interactions and their associated instabilities. The same construction can be applied to other probes, including strain and rotation, and to collective modes such as zero sound. Disorder and interband processes provide further corrections beyond the present isolated-band treatment. More broadly, the BL-FL provides a low-energy framework in which quasiparticle occupations, Landau interactions, and quantum geometry enter on an equal footing. This becomes particularly important in narrow and flat bands, where the conventional band velocity ceases to be the dominant scale for transport and interaction-induced QG currents can take over.

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END MATTER

Generic probes and QG currents.— A slow probe or weak perturbation introduces two quantum-geometric objects: an anomalous potential $\dot{\mathbf{k}} \cdot \mathbf{a}(\mathbf{k})$ and a corresponding interaction-induced QG current. Table I sum-

marizes the forces, anomalous couplings, current vertices, and representative responses.

To treat these probes in a single treatment, let $\lambda_{\mathbf{k}}$ denote the generalized single-particle ‘charge’ carried by a quasiparticle. In the presence of the conjugate vector potential $\mathbf{A}^\lambda$, the momentum carried by each fermionic state is shifted according to $\mathbf{k} \rightarrow \mathbf{k} + \lambda_{\mathbf{k}} \mathbf{A}^\lambda$. For charge transport $\lambda_{\mathbf{k}} = e$; for heat transport $\lambda_{\mathbf{k}} = \epsilon_{\mathbf{k}}^{\text{qp}} - \mu$; and for a momentum current $\lambda_{\mathbf{k}} = k_j$. Expanding the projected interaction to linear order in $\mathbf{A}^\lambda$ gives the interaction-induced vertex

$$\mathcal{J}_\alpha^{\lambda, \text{int}}(\mathbf{k}) = \frac{1}{V} \sum_{\mathbf{k}'} [(\lambda_{\mathbf{k}} \partial_{k_\alpha} + \lambda_{\mathbf{k}'} \partial_{k'_\alpha}) f_{\mathbf{k}\mathbf{k}'}] n_{\mathbf{k}'}, \quad (15)$$

which separates naturally into center-of-mass and relative-momentum derivatives,

$$\mathcal{J}_\alpha^\lambda(\mathbf{k}) = \frac{1}{V} \sum_{\mathbf{k}'} n_{\mathbf{k}'} \left(\frac{\lambda_{\mathbf{k}} + \lambda_{\mathbf{k}'}}{2} \mathfrak{D}_\alpha + \frac{\lambda_{\mathbf{k}} - \lambda_{\mathbf{k}'}}{2} \mathfrak{d}_\alpha \right) f_{\mathbf{k}\mathbf{k}'}, \quad (16)$$

where $\mathfrak{D}_\alpha = \partial_{k_\alpha} + \partial_{k'_\alpha}$ and $\mathfrak{d}_\alpha = \partial_{k_\alpha} - \partial_{k'_\alpha}$ act only on the interaction kernel. The $\mathfrak{D}_\alpha$ term probes the center-of-mass momentum dependence of the projected interaction. By contrast, the $\mathfrak{d}_\alpha$ term describes the conventional relative-momentum, or backflow, channel. For charge transport, $\lambda_{\mathbf{k}} = e$ is momentum independent, so the relative-momentum term vanishes identically and Eq. (16) reduces to $\gamma^{(\text{int}), \omega}$ of Eq. (7). The derivations and descriptions for the probes listed in Table I are given in the Supplemental Material [68].

Numerical method.— All data in Fig. 1 are obtained from self-consistent finite-temperature HF calculations on a 121×121 momentum grid (221×221 for the flat-band case). Temperature is swept from high to low, with each converged solution used to initialize the next, until the HF residual falls below 2×10^{-12} . We define the Fermi scale of each branch as $T_F = \mu - \min_{\mathbf{k}} \epsilon_{\mathbf{k}}^{\text{qp}}$. The full AHE in Fig. 1(a) is evaluated from the many-body Berry curvature of the self-consistent HF state under boundary twists. At each temperature, we first determine the equilibrium occupations $n_{\mathbf{k}}^0$ from the HF approximation for the system with periodic boundary conditions. The occupations are then kept fixed while HF mean-field parameters are self-consistently reconverged under the boundary twists $(\theta_x, \theta_y) = (\pm \delta\theta/2, \pm \delta\theta/2)$, with $\delta\theta = 0.1$. Denoting the resulting upper-band projectors by $P_{1,\dots,4}(\mathbf{k})$, the gauge-invariant Berry flux associated with each momentum label is $\phi_{\mathbf{k}}^{\text{full}} = \arg \text{Tr} [P_1(\mathbf{k}) P_2(\mathbf{k}) P_3(\mathbf{k}) P_4(\mathbf{k})]$. The Hall conductivity is then

$$\sigma_{xy}^{\text{full}} = \frac{e^2}{h} \frac{2\pi}{\delta\theta^2} \sum_{\mathbf{k}} n_{\mathbf{k}}^{(0)} \phi_{\mathbf{k}}^{\text{full}}. \quad (17)$$

In turn, σ_{xy}^{qp} associated with the quasiparticle band is obtained from this formula with the projectors under twisted boundary conditions constructed using the HF

mean-field parameters determined under periodic boundary conditions.

The Kohn stiffness in Fig. 1(c) is extracted from the curvature of the free energy with respect to a boundary twist $D_{xx} = \pi e^2 \partial_{A_x}^2 F_{\text{HF}}(A_x)|_{A_x \rightarrow 0}$, where A_x is the uniform vector potential corresponding to the twist in the x direction. The curvature is evaluated using a fourth-order five-point stencil. The result agrees with the independent linear-response calculation to relative accuracy 7×10^{-5} for all branches and temperatures.

Drude weight at finite temperature.— The temperature dependence of the Drude weight can be understood from its energy-resolved transport weight,

$$D_{xx}(T) = \pi e \int d\varepsilon \mathcal{N}_\Phi(\varepsilon) \left(-\frac{\partial n_F}{\partial \varepsilon} \right), \quad (18)$$

where $\mathcal{N}_\Phi(\varepsilon) = \int_{\mathbf{k}} \delta(\varepsilon - \epsilon_{\mathbf{k}}^{\text{qp}}) v_x^{\text{qp}}(\mathbf{k}) \mathcal{J}_{L,x}(\mathbf{k})$. For the Wilson-Dirac model in the flat-band limit, $\mathcal{N}_\Phi \propto k^2$ near the band minimum, where the QG current is weak. Thermal broadening can yield three regimes as the temperature increases. At low temperature, we obtain the Sommerfeld behavior $D(T) \simeq D(0) [1 + a(T/T_F)^2]$ with $D(0) \propto k_F^2$. For $T \gtrsim T_F$, the contribution from states within the energy window $\Delta\varepsilon$ around the Fermi energy gives $D(T) \simeq \frac{A}{k_B T} e^{-\Delta\varepsilon/k_B T}$, where $\Delta\varepsilon \sim k_B T_F$ is the characteristic excitation energy of the states dominating transport. This implies a maximum at $k_B T^* \sim \Delta_B$ in $D(T)$. At even higher temperature, the occupations become nearly uniform and the Drude weight crosses over to the $1/T$ decay as discussed in the main text.

TABLE I. Generic forces, anomalous potential, and interaction-induced QG currents in a BL-FL theory. A perturbation that drives the quasiparticle momentum according to $\dot{\mathbf{k}}$ generates the anomalous potential $\dot{\mathbf{k}} \cdot \mathbf{a}(\mathbf{k})$ where $\mathbf{a}(\mathbf{k})$ is, respectively, the Berry connection of the quasiparticle (bare particle) band in an interacting (non-interacting) system. For a probe coupling to a quasiparticle with generalized charge $\lambda_{\mathbf{k}}$, the interaction-induced QG current is given in Eq. (16) in the BL-FL theory. In the table, as detailed in the Supplemental Material [68], $\varphi(\mathbf{r})$ is a smooth single-particle potential energy; u_{ij} denotes strain with elastic gauge potentials $\mathbf{A}_{\text{el}}$ and Φ_{el} ; ω_z is the local frame-rotation rate; and $s_z^{\mathbf{k}} = \langle u_{\mathbf{k}} | S_z^{\text{orbital}} | u_{\mathbf{k}} \rangle$ is the projection of the orbital rotation generator onto the quasiparticle band. Furthermore, $\Gamma_{ij} = \dot{e}_i^a e_{ja}$ whose symmetric and antisymmetric parts describe strain and frame rotation, respectively; and T_{0i}^a denotes torsion generated by a time-dependent vielbein gradient.

	Generalized force $\dot{\mathbf{k}}$	Anomalous potential	Charge $\lambda_{\mathbf{k}}$ in QG current	Representative response
Electric field	$-e\mathbf{E}$	$-e\mathbf{E} \cdot \mathbf{a}(\mathbf{k})$	e	AHE [8, 43] electric polarization [71] Drude weight
Magnetic field	$-e\dot{\mathbf{r}} \times \mathbf{B}$, $\dot{\mathbf{r}} = \nabla_{\mathbf{k}} \epsilon_{\mathbf{k}}^{\text{qp}}$	$-e(\dot{\mathbf{r}} \times \mathbf{B}) \cdot \mathbf{a}(\mathbf{k})$	e	magnetization [10, 72], Streda response [73], cyclotron dynamics
Scalar potential	$-\nabla_{\mathbf{r}} \varphi(\mathbf{r})$	$-\nabla \varphi \cdot \mathbf{a}(\mathbf{k})$	e	AHE
Thermal	$(\epsilon_{\mathbf{k}}^{\text{qp}} - \mu) \frac{\nabla T}{T}$	$(\epsilon_{\mathbf{k}}^{\text{qp}} - \mu) \frac{\nabla T}{T} \cdot \mathbf{a}(\mathbf{k})$	$\epsilon_{\mathbf{k}}^{\text{qp}} - \mu$	thermal Hall [67, 74–76], thermoelectricity
Strain acoustic field	$-\partial_t \mathbf{A}_{\text{el}} - \nabla \Phi_{\text{el}}$, or locally $-\dot{u}_{ij} k_j$	$(-\partial_t \mathbf{A}_{\text{el}} - \nabla \Phi_{\text{el}}) \cdot \mathbf{a}(\mathbf{k})$	k_j	piezoelectric [71, 77], elastoconductivity [78], phonon drag [79, 80]
Frame rotation spin connection	$-\omega_z \hat{\mathbf{z}} \times \mathbf{k}$ with $D_t = \partial_t - i\omega_z S_z^{\text{basis}}$	$-\omega_z (\hat{\mathbf{z}} \times \mathbf{k}) \cdot \mathbf{a} + \omega_z s_z(\mathbf{k})$	$(\hat{\mathbf{z}} \times \mathbf{k})_j$	orbital spin, Hall viscosity [81–83], rotational susceptibility
Metric or shear quench	$-\Gamma_{ij} k_j$	$-\Gamma_{ij} k_j a_i(\mathbf{k})$	k_j	nematic susceptibility [84, 85], dynamical Hall viscosity in optical response [86, 87]
Torsion/vielbein gradient	$-T_{0i}^a k_a$ in the local frame	$-T_{0i}^a k_a a_i(\mathbf{k})$	$(\epsilon_{\mathbf{k}}^{\text{qp}} - \mu) k_a$	torsional viscosity [88], energy magnetization [74]

Supplemental Material for “Berry-Landau Fermi-liquid theory: transport in presence of quantum geometry”

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S1. GENERALIZED PEIERLS SUBSTITUTION

A. Grassmann-valued Lagrange-multiplier projection

For completeness and consistency, we revisit the generalized Peierls substitution approach introduced in Ref. [25] for a non-interacting system. We consider a system with an active band separated from the remote bands by a large energy gap. The band structure is described by N_{orb} -orbital Hamiltonian $H_0 = \sum_{\mathbf{k}} h(\mathbf{k})$ with

$$h(\mathbf{k}) = \sum_{\alpha\beta} c_{\mathbf{k}\alpha}^\dagger t_{\alpha\beta}(\mathbf{k}) c_{\mathbf{k}\beta}. \quad (\text{S1})$$

Here $c_{\mathbf{k}\alpha}^\dagger$ creates a fermion on orbital α at momentum $\mathbf{k}$. In particular, we denote the Bloch wave at the crystal momentum $\mathbf{k}$ as $|u_{\mathbf{k}}\rangle$: $h(\mathbf{k})|u_{\mathbf{k}}\rangle = \epsilon_{\mathbf{k}}|u_{\mathbf{k}}\rangle$. We can project the whole system onto the Hilbert subspace $\mathbb{H}^{\text{sub}}$ of the active band. According to Ref. [25], the band projection can be realized by means of the Grassmann-valued Lagrange multipliers $\lambda_{\mathbf{k}\alpha}$ and $\lambda_{\mathbf{k}\alpha}^\dagger$

$$S_\lambda = \int_0^\beta d\tau \sum_{\mathbf{k}\alpha} \left[\lambda_{\mathbf{k}\alpha}^\dagger(\tau) (c_{\mathbf{k}\alpha}(\tau) - u_{\mathbf{k}\alpha}(\tau) f_{\mathbf{k}}(\tau)) + (c_{\mathbf{k}\alpha}^\dagger(\tau) - u_{\mathbf{k}\alpha}^*(\tau) f_{\mathbf{k}}^\dagger(\tau)) \lambda_{\mathbf{k}\alpha}(\tau) \right], \quad (\text{S2})$$

where the operator $f_{\mathbf{k}}$ annihilates a fermion at the momentum $\mathbf{k}$ with the Bloch wave $|u_{\mathbf{k}}\rangle$. The partition function in imaginary time can be formulated as

$$Z_0 = \int Dc^\dagger Dc Df^\dagger Df D\lambda^\dagger D\lambda e^{-S_0 - S_\lambda}, \quad (\text{S3})$$

where $S_0 = \int d\tau \left[\sum_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha}^\dagger(\tau) \partial_\tau c_{\mathbf{k}\alpha}(\tau) + H_0 \right]$. We can obtain the band projection $c_{\mathbf{k}\alpha} = u_{\mathbf{k}\alpha} f_{\mathbf{k}}$ by integrating out the Lagrange multipliers $\lambda_{\mathbf{k}\alpha}$ and $\lambda_{\mathbf{k}\alpha}^\dagger$. Within the Hilbert subspace $\mathbb{H}^{\text{sub}}$, all fields λ , f and c carry $U(1)$ charge. The band projector is modified in the presence of an electromagnetic field. For instance, the Bloch wave becomes time dependent as its crystal momentum follows the semiclassical dynamical equation (see Sec. S1C).

B. Band projection under interactions

We show here how the Bloch wave entering the band-projection constraint is determined in the interacting theory. We first formulate the Baym-Kadanoff functional for the full single-particle Green function and then restrict it to the low-energy quasiparticle subspace through the constraint S_λ .

For convenience, we use $i = (\tau_i, \mathbf{k}_i, \alpha_i)$ to collect the imaginary time, momentum, and orbital indices. The Euclidean action is given by $S[c^\dagger, c] = -c_1^\dagger (G_0^{-1})_{12} c_2 + S_{\text{int}}[c^\dagger, c]$, where repeated indices are summed or integrated over. Here G_0 is the bare Green function with

$$G_0^{-1}(i\omega_n, \mathbf{k}) = i\omega_n + \mu - h_0(\mathbf{k}). \quad (\text{S4})$$

We introduce the source term J_{12} to the partition function,

$$Z[J] = \int D[c^\dagger, c] \exp \left[-S[c^\dagger, c] - c_1^\dagger J_{12} c_2 \right]. \quad (\text{S5})$$

Formally, we can obtain the Green function $G_{12} = \frac{\delta W}{\delta J_{12}}$, with $W[J] = -\ln Z[J]$. The Baym-Kadanoff functional $\Gamma[G]$ is obtained from the Legendre transformation

$$\Gamma[G] = W[J] - \text{Tr}(JG), \quad (\text{S6})$$

which satisfies $\frac{\delta \Gamma}{\delta G} = -J$. The full single-particle Green function corresponds to the stationary point $\frac{\delta \Gamma[G]}{\delta G} = 0$. For the non-interacting system, we have $G^{-1} = G_0^{-1} - J$. From

$$Z_0[J] = \int D[c^\dagger, c] \exp \left[-c^\dagger (G_0^{-1} - J) c \right] \propto \det [G_0^{-1} - J], \quad (\text{S7})$$

we obtain

$$\Gamma_0[G] = -\ln Z_0[J] - \text{Tr}(JG) = \text{Tr} \ln G - \text{Tr} [(G_0^{-1} - G^{-1}) G]. \quad (\text{S8})$$

Interactions add the Luttinger-Ward functional $\Phi[G]$, giving

$$\Gamma[G] = \text{Tr} \ln G - \text{Tr} [(G_0^{-1} - G^{-1}) G] + \Phi[G]. \quad (\text{S9})$$

Here $\Phi[G]$ is the sum of closed two-particle-irreducible diagrams constructed from the full Green function G and bare interaction vertices. Equivalently, $\Phi[G]$ may be defined non-perturbatively as the interaction part of the two-particle-irreducible effective action. The self-energy can be deduced via $\Sigma_{12} = \frac{\delta \Phi[G]}{\delta G_{12}}$.

We impose the constraint S_λ on the Baym-Kadanoff functional $\Gamma[G]$ by integrating over the Lagrange multipliers $\lambda^\dagger$ and λ . Writing $k = (i\omega_n, \mathbf{k})$, we decompose the full Green function

$$G(k) = G_P(k) + G_Q(k), \quad (\text{S10})$$

with $G_P = PGP$ and $G_Q = G - G_P$, where P denotes the band projector onto the active band. The intraband component G_P possesses the structure $G_P(k) = g(k)P_{\mathbf{k}}$ with $g(k) = \langle f_k f_k^\dagger \rangle$ and $P_{\mathbf{k}} = |u_{\mathbf{k}}\rangle \langle u_{\mathbf{k}}|$. Physically, G_P (G_Q) is the coherent (incoherent) part of the Green function. In the isolated-band limit, G_Q vanishes and thus $G(k) = G_P(k)$.

We restrict the Baym-Kadanoff functional to the Hilbert subspace,

$$\tilde{\Gamma}[g, P] \equiv \Gamma(G = gP). \quad (\text{S11})$$

The constraint S_λ does not provide an additional contribution to $\Phi[G]$. After the band projection, the Luttinger-Ward functional $\Phi[G]$ becomes $\Phi_{\text{proj}}[g, P] = \Phi[G = gP]$. We are now ready to determine the parameters $g(k)$ and $P_{\mathbf{k}}$. First, we vary the scalar propagator $g(k)$ while fixing $P_{\mathbf{k}}$. From $\delta G(k) = P_{\mathbf{k}}\delta g(k)$ and $G^{-1} = g^{-1}P + G_Q^{-1}$, we have the variation

$$\delta\Gamma = \sum_k \delta g(k) \text{Tr} [P(G^{-1} - (G_0^{-1} - \Sigma))] = 0. \quad (\text{S12})$$

Given the isolated-band condition, Eq. (S12) reduces to

$$\frac{\delta\tilde{\Gamma}}{\delta g(k)} = g^{-1}(k) - \langle u_{\mathbf{k}} | (G_0^{-1} - \Sigma) | u_{\mathbf{k}} \rangle = 0. \quad (\text{S13})$$

Thus, we obtain the scalar quasiparticle propagator $g^{-1}(k) = \langle u_{\mathbf{k}} | G^{-1} | u_{\mathbf{k}} \rangle$. Second, we consider the variation of the Bloch wave $u_{\mathbf{k}}$. The normalization $\langle u_{\mathbf{k}} | u_{\mathbf{k}} \rangle = 1$ indicates that the physically relevant variation is perpendicular to $u_{\mathbf{k}}$. Thus, we can represent the variation as an infinitesimal rotation $P_{\mathbf{k}} \rightarrow e^{\eta_{\mathbf{k}}} P_{\mathbf{k}} e^{-\eta_{\mathbf{k}}}$, which gives $\delta P_{\mathbf{k}} = [\eta_{\mathbf{k}}, P_{\mathbf{k}}]$. Equivalently, we can write $|\delta u_{\mathbf{k}}\rangle = Q_{\mathbf{k}} \eta_{\mathbf{k}} | u_{\mathbf{k}} \rangle$ up to an irrelevant change of the phase of $u_{\mathbf{k}}$. We apply the same infinitesimal transformation to Eq. (S10), $G \rightarrow e^\eta G e^{-\eta}$, or equivalently, $\delta_u G = [\eta, G]$. Therefore, we have

$$\begin{aligned} \delta_u \Gamma &= \text{Tr} [(G^{-1} - (G_0^{-1} - \Sigma)) [\eta, G]] \\ &= -\text{Tr} [\eta [G, G_0^{-1} - \Sigma]] \end{aligned} \quad (\text{S14})$$

where we used the identity $\text{Tr}[G^{-1}[\eta, G]] = \text{Tr}[G^{-1}\eta G - G^{-1}G\eta] = 0$. Stationarity with respect to arbitrary orbital rotations η gives $[G, G_0^{-1} - \Sigma] = 0$. From $G(k)|u_{\mathbf{k}}\rangle = g(k)P_{\mathbf{k}}|u_{\mathbf{k}}\rangle = g(k)|u_{\mathbf{k}}\rangle$, we obtain the eigenvalue problem,

$$G(G_0^{-1} - \Sigma)|u_{\mathbf{k}}\rangle = (G_0^{-1} - \Sigma)G|u_{\mathbf{k}}\rangle = g(k)(G_0^{-1} - \Sigma)|u_{\mathbf{k}}\rangle. \quad (\text{S15})$$

This implies that $(G_0^{-1} - \Sigma)|u_{\mathbf{k}}\rangle$ is also an eigenvector of G with eigenvalue $g(k)$. Due to the non-degeneracy of the quasiparticle eigenvalues, we further simplify the eigenvalue problem

$$(G_0^{-1} - \Sigma)|u_{\mathbf{k}}\rangle = g^{-1}(k)|u_{\mathbf{k}}\rangle. \quad (\text{S16})$$

This gives the condition for Bloch waves in the presence of interactions. Let $z_{\mathbf{k}}$ denote the quasiparticle pole, $g^{-1}(z_{\mathbf{k}}, \mathbf{k}) = 0$. We arrive at the condition used in the main text, $G^{-1}(z_{\mathbf{k}}, \mathbf{k})|u_{\mathbf{k}}\rangle = 0$, with $G^{-1} = G_0^{-1} - \Sigma$. For a long-lived quasiparticle, $z_{\mathbf{k}} = \xi_{\mathbf{k}}^{\text{qp}} - i\Gamma_{\mathbf{k}}$ with $\Gamma_{\mathbf{k}} \rightarrow 0$, the anti-Hermitian part of the self-energy Σ can be neglected to leading order.

The role of the remote bands can be shown explicitly. Writing the inverse Green function in the block form,

$$G^{-1} = \begin{pmatrix} K_{PP} & K_{PQ} \\ K_{QP} & K_{QQ} \end{pmatrix}, \quad |u^{\text{qp}}\rangle = \begin{pmatrix} u_P \\ u_Q \end{pmatrix}, \quad (\text{S17})$$

the equation $G^{-1}(z_{\mathbf{k}}, \mathbf{k})|u_{\mathbf{k}}\rangle = 0$ leads to $K_{QP}u_P + K_{QQ}u_Q = 0$, and thus $u_Q = -K_{QQ}^{-1}K_{QP}u_P$. With the gap Δ between the active and remote bands, we have $K_{QQ}^{-1} \sim \frac{1}{\Delta}$ and $K_{QP} \sim U$ with U the interaction scale, so that $u_Q \sim \frac{U}{\Delta}u_P$. Thus, interactions allow hybridization between the active and remote bands; the hybridization is small only when $U/\Delta \ll 1$. Eliminating u_Q gives the exact active-sector pole equation $[K_{PP} - K_{PQ}K_{QQ}^{-1}K_{QP}]u_P = 0$.

C. Houston Ansatz for the Bloch wave on an accelerated trajectory

We consider how the band projector in Eq. (S2) changes in response to a weak static electric field. We choose the velocity gauge $\mathbf{A}(t) = -\mathbf{E}t$ for the static electric field (here t is real time). Then, we have the Hamiltonian $H_0(\mathbf{k}, t) = H_0(\mathbf{k} + e\mathbf{A}(t))$. The Houston Ansatz tells us that the solution to the instantaneous Schrödinger equation $H_0(\mathbf{k} + e\mathbf{A}(t))|u_{n\mathbf{k}}(t)\rangle = \epsilon_n(\mathbf{k}, t)|u_{n\mathbf{k}}(t)\rangle$ (with n the band index) takes the form $|u_{n\mathbf{k}}(t)\rangle = |u_{n\mathbf{k}(t)}\rangle$, where the crystal momentum $\mathbf{k}$ follows the classical dynamics $\dot{\mathbf{k}}(t) = -e\mathbf{E}$.

For convenience, we restate the Houston Ansatz in imaginary time. After the Wick rotation $t = -i\tau$, the velocity gauge becomes $\mathbf{A}(\tau) = -(-i\tau)\mathbf{E} = i\mathbf{E}\tau$. Therefore, the imaginary-time Hamiltonian in the velocity gauge is $H_E(\tau) = H_0(\mathbf{k} + ie\mathbf{E}\tau)$. Accordingly, we have the imaginary-time Houston Ansatz $|u_{n\mathbf{k}}(\tau)\rangle = |u_{n\mathbf{k}(\tau)}\rangle$ with the crystal momentum following the evolution equation $d\mathbf{k}/d\tau = -e\mathbf{E}$.

D. Gravitomagnetic Peierls substitution

To study thermal transport, we couple the system to a background gravitomagnetic vector potential $\boldsymbol{\lambda}$.

1. Energy twist from the gravitomagnetic field

We place the system on a Euclidean torus with the metric $ds^2 = (d\tau + i\lambda_i dx^i)^2 + \delta_{ij} dx^i dx^j$. Here the off-diagonal metric component $g_{0i} = i\lambda_i$ is a gravitomagnetic vector potential, conjugate to the energy current. The gravitomagnetic background appears through the global identifications of the Euclidean torus,

$$(\tau, \mathbf{x}) \sim (\tau + \beta, \mathbf{x}) \sim (\tau + i\lambda_i L_i, \mathbf{x} + L_i \mathbf{e}_i), \quad (\text{S18})$$

with $\mathbf{e}_i$ the unit vector along the x^i -direction. The introduction of the gravitomagnetic field twists the boundary condition. This is the gravitational analogue of a flux-induced boundary twist.

In imaginary time, the Bloch wave is written as $|\psi_{\mathbf{k}}\rangle = e^{-\epsilon(\mathbf{k})\tau} e^{i\mathbf{k}\cdot\mathbf{x}} |u_{\mathbf{k}}\rangle$. Under the gravitomagnetic twisted boundary condition $(\tau, \mathbf{x}) \sim (\tau + i\lambda_i L_i, \mathbf{x} + L_i \mathbf{e}_i)$, the state $|\psi_{\mathbf{k}}\rangle$ acquires a factor $e^{-i\lambda_i L_i \epsilon(\mathbf{k})} e^{iL_i k_i}$ after winding once around the torus. Single-valuedness gives the quantization condition for the momentum $\mathbf{k}$

$$(k_i - \epsilon(\mathbf{k})\lambda_i)L_i = 2\pi n_i. \quad (\text{S19})$$

We define the twisted momentum $\mathbf{k}^\lambda = \mathbf{k} + \epsilon(\mathbf{k}^\lambda)\boldsymbol{\lambda}$; the corresponding energy is $\epsilon^\lambda(\mathbf{k}) = \epsilon(\mathbf{k}^\lambda) = \epsilon(\mathbf{k} + \boldsymbol{\lambda}\epsilon(\mathbf{k}^\lambda))$. This is the gravitomagnetic analogue of the usual Peierls substitution. To linear order in $\boldsymbol{\lambda}$, $\mathbf{k}^\lambda = \mathbf{k} + \boldsymbol{\lambda}\epsilon(\mathbf{k}) + \mathcal{O}(\lambda^2)$, such that $\mathbf{k} \rightarrow \mathbf{k} + \epsilon(\mathbf{k})\boldsymbol{\lambda}$. We will derive the quantum-geometric thermal current through the introduction of the gravitomagnetic vector potential.

2. Quantum-geometric thermal current

To derive the quantum-geometric thermal current, we solve the dynamics of the crystal momentum. We consider the Lagrangian for the gravitomagnetic coupling with $\boldsymbol{\lambda} = \boldsymbol{\lambda}(t)$: $L = (\mathbf{k} + \epsilon^\lambda(\mathbf{k})\boldsymbol{\lambda}) \cdot \dot{\mathbf{r}} - \epsilon^\lambda(\mathbf{k})$ with the canonical momentum $\boldsymbol{\pi} = \frac{\partial L}{\partial \dot{\mathbf{r}}} = \mathbf{k} + \epsilon^\lambda(\mathbf{k})\boldsymbol{\lambda}$. From the Euler-Lagrange equation $\frac{d}{dt}(\mathbf{k} + \epsilon^\lambda(\mathbf{k})\boldsymbol{\lambda}) = 0$, the crystal momentum $\mathbf{k}$ follows the equation of motion $d_t \mathbf{k} = -\epsilon(\mathbf{k})d_t \boldsymbol{\lambda}$ to first order in $\boldsymbol{\lambda}$. This gives the overlap

$$\begin{aligned} -i\langle u_{\mathbf{k}} | \partial_\tau u_{\mathbf{k}} \rangle &= d_t \mathbf{k} \cdot \mathbf{a}(\mathbf{k}) \\ &= -\epsilon_{\mathbf{k}} d_t \boldsymbol{\lambda} \cdot \mathbf{a}(\mathbf{k}) \equiv \epsilon_{\mathbf{k}} \mathbf{E}^g \cdot \mathbf{a}(\mathbf{k}), \end{aligned} \quad (\text{S20})$$

where the gravitoelectric field $\mathbf{E}^g = -d_t \boldsymbol{\lambda}$ plays the role of the temperature gradient. We next determine the first-order perturbation to the Bloch wave due to the gravitomagnetic field. To linear order in $\boldsymbol{\lambda}$, the gravitomagnetic Peierls substitution gives $H^\lambda(\mathbf{k}) = H(\mathbf{k}) + \sum_\alpha \lambda_\alpha j_\alpha^E(\mathbf{k}) + \mathcal{O}(\lambda^2)$ with the energy-current operator $j_\alpha^E = \frac{1}{2} \{\partial_\alpha H(\mathbf{k}), H(\mathbf{k})\}$. Expanding the Bloch wave as $|u_{\mathbf{k}}^\lambda\rangle = |u_{\mathbf{k}}\rangle + \lambda_\alpha |\delta_\alpha^\lambda u_{\mathbf{k}}\rangle + \mathcal{O}(\lambda^2)$, perturbation theory gives

$$|\delta_\alpha^\lambda u_{\mathbf{k}}\rangle = \sum_m |u_{m\mathbf{k}}\rangle \frac{\langle u_{m\mathbf{k}} | j_\alpha^E(\mathbf{k}) | u_{\mathbf{k}} \rangle}{\epsilon_{\mathbf{k}} - \epsilon_{m\mathbf{k}}} = \frac{\epsilon_{\mathbf{k}} + H(\mathbf{k})}{2} (1 - |u_{\mathbf{k}}\rangle \langle u_{\mathbf{k}}|) |\partial_\alpha u_{\mathbf{k}}\rangle, \quad (\text{S21})$$

with $\langle u_{m\mathbf{k}} | j_\alpha^E(\mathbf{k}) | u_{\mathbf{k}} \rangle = \frac{1}{2} (\epsilon_{\mathbf{k}}^2 - \epsilon_{m\mathbf{k}}^2) \langle u_{m\mathbf{k}} | \partial_\alpha u_{\mathbf{k}} \rangle$. We can then construct the corresponding terms in the energy functional of the BL-FL theory. Explicitly, we have

$$\delta E[n, \lambda] = \sum_{\mathbf{k}} \xi_{\mathbf{k}}^{\text{qp}} \mathbf{E}^g \cdot \mathbf{a}(\mathbf{k}) + \frac{1}{2V} \sum_{\alpha} \lambda_{\alpha} (\mathcal{J}_{\alpha}^{Q, \text{QG}} + \mathcal{J}_{\alpha}^{Q, \text{backflow}}), \quad (\text{S22})$$

with

$$\mathcal{J}_{\alpha}^{Q, \text{QG}} = \frac{1}{4V} \sum_{\mathbf{k}\mathbf{k}'} (\xi_{\mathbf{k}} + \xi_{\mathbf{k}'}) (\partial_{k_{\alpha}} + \partial_{k'_{\alpha}}) f_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}} n_{\mathbf{k}'}, \quad (\text{S23})$$

$$\mathcal{J}_{\alpha}^{Q, \text{backflow}} = \frac{1}{4V} \sum_{\mathbf{k}\mathbf{k}'} (\xi_{\mathbf{k}} - \xi_{\mathbf{k}'}) (\partial_{k_{\alpha}} - \partial_{k'_{\alpha}}) f_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}} n_{\mathbf{k}'}. \quad (\text{S24})$$

E. Quantum-geometric current and Hartree-Fock approximation

We apply the Hartree-Fock approximation to the interactions and to the quantum-geometric current operator. Starting from the density-density interaction

$$H_{\text{int}} = \frac{1}{2V} \sum_{\mathbf{p}\mathbf{p}'\mathbf{q}\alpha\beta} V_{\mathbf{q}} c_{\mathbf{p}+\mathbf{q}\alpha}^\dagger c_{\mathbf{p}'-\mathbf{q}\beta}^\dagger c_{\mathbf{p}'\beta} c_{\mathbf{p}\alpha}, \quad (\text{S25})$$

the Hartree-Fock approximation gives an HF Hamiltonian $H_0^{\text{HF}}(\mathbf{k}) = \sum_{\alpha\beta} t_{\alpha\beta}^{\text{HF}}(\mathbf{k}) c_{\mathbf{k}\alpha}^\dagger c_{\mathbf{k}\beta}$ with

$$t_{\alpha\beta}^{\text{HF}}(\mathbf{k}) = t_{\alpha\beta}(\mathbf{k}) - \frac{1}{V} \sum_{\mathbf{k}'} V_{\mathbf{k}-\mathbf{k}'} \langle c_{\mathbf{k}'\alpha}^\dagger c_{\mathbf{k}'\beta} \rangle.$$

$H_0^{\text{HF}}(\mathbf{k})$ defines the Bloch states of the quasiparticles in the active HF band:

$$H_0^{\text{HF}}(\mathbf{k}) |u_{\mathbf{k}}^{\text{HF}}\rangle = \epsilon_{\mathbf{k}}^{\text{HF}} |u_{\mathbf{k}}^{\text{HF}}\rangle, \quad (\text{S26})$$

$$\mathbf{a}^{\text{HF}}(\mathbf{k}) = -i \langle u_{\mathbf{k}}^{\text{HF}} | \nabla_{\mathbf{k}} | u_{\mathbf{k}}^{\text{HF}} \rangle. \quad (\text{S27})$$

After projection onto the active HF band, the interaction term becomes,

$$H_{\text{int}} = \frac{1}{2V} \sum_{\mathbf{p}\mathbf{p}'\mathbf{q}} V_{\mathbf{q}} \langle u_{\mathbf{p}+\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}}^{\text{HF}} \rangle \langle u_{\mathbf{p}'-\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle f_{\mathbf{p}+\mathbf{q}}^\dagger f_{\mathbf{p}'-\mathbf{q}}^\dagger f_{\mathbf{p}'} f_{\mathbf{p}}. \quad (\text{S28})$$

We introduce a weak static electric field through the gauge potential $\mathbf{A}(\tau) = i\mathbf{E}\tau$. Within the Houston Ansatz, the Hartree-Fock Bloch wave follows $|u_{\mathbf{p}}^{\text{HF}}(\tau)\rangle = |u_{\mathbf{p}(\tau)}^{\text{HF}}\rangle$ with $\dot{\mathbf{p}}(\tau) = -e\mathbf{E}$. After integrating out the Lagrange multipliers $\lambda, \lambda^\dagger$ and the fermion fields $c, c^\dagger$, the interaction term becomes coupled to the gauge potential; to leading order in $\mathbf{A}$,

$$H_{\text{int}}[A] = H_{\text{int}} + \frac{1}{2V} \sum_{\mathbf{p}\mathbf{p}'\mathbf{q}} \int_{\omega} e A_{\alpha}(\omega) e^{-i\omega\tau} V_{\mathbf{q}} \mathfrak{D}_{\alpha} [\langle u_{\mathbf{p}+\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}}^{\text{HF}} \rangle \langle u_{\mathbf{p}'-\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle] f_{\mathbf{p}+\mathbf{q}}^\dagger f_{\mathbf{p}'-\mathbf{q}}^\dagger f_{\mathbf{p}'} f_{\mathbf{p}} + \mathcal{O}(A^2). \quad (\text{S29})$$

Here we define a derivative operator $\mathfrak{D}_{\alpha}$, satisfying

$$\mathfrak{D}_{\alpha} (\langle u_{\mathbf{p}+\mathbf{q}} | u_{\mathbf{p}} \rangle) = \langle \mathcal{D}_{\alpha} u_{\mathbf{p}+\mathbf{q}} | u_{\mathbf{p}} \rangle + \langle u_{\mathbf{p}+\mathbf{q}} | \mathcal{D}_{\alpha} u_{\mathbf{p}} \rangle, \quad (\text{S30})$$

$$\mathfrak{D}_{\alpha} [\langle u_{\mathbf{p}+\mathbf{q}} | u_{\mathbf{p}} \rangle \langle u_{\mathbf{p}'-\mathbf{q}} | u_{\mathbf{p}'} \rangle] = \mathfrak{D}_{\alpha} (\langle u_{\mathbf{p}+\mathbf{q}} | u_{\mathbf{p}} \rangle) \langle u_{\mathbf{p}'-\mathbf{q}} | u_{\mathbf{p}'} \rangle + \langle u_{\mathbf{p}+\mathbf{q}} | u_{\mathbf{p}} \rangle \mathfrak{D}_{\alpha} (\langle u_{\mathbf{p}'-\mathbf{q}} | u_{\mathbf{p}'} \rangle), \quad (\text{S31})$$

with $\mathcal{D}_{\alpha}$ the covariant derivative $|\mathcal{D}_{\alpha} u_{\mathbf{p}}\rangle = (\partial_{p_{\alpha}} - i a_{\alpha}(\mathbf{p})) |u_{\mathbf{p}}\rangle$. From Eq. (S29), we can identify the quantum-geometric current operator

$$J_{\alpha}^{\text{QG}}(\mathbf{p}) = \frac{1}{V} \sum_{\mathbf{p}'\mathbf{q}} V_{\mathbf{q}} \mathfrak{D}_{\alpha} [\langle u_{\mathbf{p}+\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}}^{\text{HF}} \rangle \langle u_{\mathbf{p}'-\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle] f_{\mathbf{p}+\mathbf{q}}^\dagger f_{\mathbf{p}'-\mathbf{q}}^\dagger f_{\mathbf{p}'} f_{\mathbf{p}}. \quad (\text{S32})$$

The operator $J_{\alpha}^{\text{QG}}(\mathbf{p})$ can be simplified within the HF approximation. Explicitly, we have

$$\begin{aligned} & \frac{1}{2} V_{\mathbf{q}} \mathfrak{D}_{\alpha} [\langle u_{\mathbf{p}+\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}}^{\text{HF}} \rangle \langle u_{\mathbf{p}'-\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle] f_{\mathbf{p}+\mathbf{q}}^\dagger f_{\mathbf{p}'-\mathbf{q}}^\dagger f_{\mathbf{p}'} f_{\mathbf{p}} \\ & \rightarrow \frac{1}{2} V_0 \mathfrak{D}_{\alpha} [\langle u_{\mathbf{p}}^{\text{HF}} | u_{\mathbf{p}}^{\text{HF}} \rangle \langle u_{\mathbf{p}'}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle] [n_{\mathbf{p}}^{\text{HF}} f_{\mathbf{p}'}^\dagger f_{\mathbf{p}'} + n_{\mathbf{p}'}^{\text{HF}} f_{\mathbf{p}}^\dagger f_{\mathbf{p}}] \\ & \quad - \frac{1}{2} V_{\mathbf{p}-\mathbf{p}'} \mathfrak{D}_{\alpha} [\langle u_{\mathbf{p}'}^{\text{HF}} | u_{\mathbf{p}}^{\text{HF}} \rangle \langle u_{\mathbf{p}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle] [n_{\mathbf{p}'}^{\text{HF}} f_{\mathbf{p}}^\dagger f_{\mathbf{p}} + n_{\mathbf{p}}^{\text{HF}} f_{\mathbf{p}'}^\dagger f_{\mathbf{p}'}] \\ & \quad + \frac{1}{2} V_{\mathbf{q}} \mathfrak{D}_{\alpha} [\langle u_{\mathbf{p}+\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}}^{\text{HF}} \rangle \langle u_{\mathbf{p}'-\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle] : f_{\mathbf{p}+\mathbf{q}}^\dagger f_{\mathbf{p}'-\mathbf{q}}^\dagger f_{\mathbf{p}'} f_{\mathbf{p}} : \\ & = -V_{\mathbf{p}-\mathbf{p}'} \mathfrak{D}_{\alpha} [\langle u_{\mathbf{p}'}^{\text{HF}} | u_{\mathbf{p}}^{\text{HF}} \rangle \langle u_{\mathbf{p}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle] n_{\mathbf{p}'}^{\text{HF}} f_{\mathbf{p}}^\dagger f_{\mathbf{p}} \\ & \quad + \frac{1}{2} V_{\mathbf{q}} \mathfrak{D}_{\alpha} [\langle u_{\mathbf{p}+\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}}^{\text{HF}} \rangle \langle u_{\mathbf{p}'-\mathbf{q}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle] : f_{\mathbf{p}+\mathbf{q}}^\dagger f_{\mathbf{p}'-\mathbf{q}}^\dagger f_{\mathbf{p}'} f_{\mathbf{p}} :, \end{aligned} \quad (\text{S33})$$

where the normal-ordered term $: f^\dagger f^\dagger f f :$ denotes the residual interactions beyond the HF approximation. Thus, $H_{\text{int}}[A]$ in Eq. (S29) simplifies to

$$H_{\text{int}}[A] = H_{\text{int}} - \frac{e}{V} \sum_{\mathbf{p}'} V_{\mathbf{p}-\mathbf{p}'} \mathfrak{D}_{\alpha} [|\langle u_{\mathbf{p}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle|^2] n_{\mathbf{p}'}^{\text{HF}} A_{\alpha}(\omega) f_{\mathbf{p}}^\dagger(p_0 + \omega) f_{\mathbf{p}}(p_0). \quad (\text{S34})$$

For the coefficients

$$V_{\mathbf{p}-\mathbf{p}'} \mathcal{D}_\alpha \left[|\langle u_{\mathbf{p}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle|^2 \right] = (\partial_{p_\alpha} + \partial_{p'_\alpha}) \left[V_{\mathbf{p}-\mathbf{p}'} |\langle u_{\mathbf{p}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle|^2 \right] = -(\partial_{p_\alpha} + \partial_{p'_\alpha}) \Sigma_{\mathbf{p}\mathbf{p}'}^{\text{HF}}, \quad (\text{S35})$$

with the self-energy $\Sigma_{\mathbf{p}\mathbf{p}'}^{\text{HF}} = -V_{\mathbf{p}-\mathbf{p}'} |\langle u_{\mathbf{p}}^{\text{HF}} | u_{\mathbf{p}'}^{\text{HF}} \rangle|^2$, we obtain a compact expression for the quantum-geometric current operator

$$J_\alpha^{\text{QG}}(\mathbf{p}) = eT \sum_{p_0} \int_{\mathbf{p}'} n_{\mathbf{p}'}^{\text{HF}} (\partial_{p_\alpha} + \partial_{p'_\alpha}) \Sigma_{\mathbf{p}\mathbf{p}'}^{\text{HF}} f_{\mathbf{p}}^\dagger(p_0 + \omega) f_{\mathbf{p}}(p_0). \quad (\text{S36})$$

S2. INTRINSIC AHE IN A NON-INTERACTING SYSTEM

In this section, we provide details of the derivation of the intrinsic AHE in the non-interacting system. Within the Houston Ansatz $|u_{\mathbf{k}}(\tau)\rangle \equiv |u_{\mathbf{k}(\tau)}\rangle$, the crystal momentum follows the semiclassical equation of motion $\dot{\mathbf{k}}(\tau) = e\dot{\mathbf{A}}(\tau) = -e\mathbf{E}(\tau)$. After imposing the band projection, we have the Lagrangian density $\mathcal{L}$

$$\mathcal{L} = f^\dagger(\mathbf{x}, \tau) [\partial_\tau + \epsilon(-i\nabla + e\mathbf{A}) - ie\mathbf{E} \cdot \mathbf{a}(-i\nabla)] f(\mathbf{x}, \tau), \quad (\text{S37})$$

where formally $\mathbf{a}(-i\nabla)$ is understood as a pseudo-differential operator by replacing $\mathbf{k} \rightarrow -i\nabla$. We regularize a uniform static electric field $\mathbf{E}$ by a finite frequency ω with $\mathbf{E}(\omega) = i\omega\mathbf{A}(\omega)$ and take the static limit $\omega \rightarrow 0$ at the end. We obtain the effective action by integrating out the fermion fields, giving

$$Z[A] = \exp(-S_{\text{eff}}[A]), \quad (\text{S38})$$

$$S_{\text{eff}}[A] = -\text{Tr} \ln(G_0^{-1} - V[A]), \quad (\text{S39})$$

with

$$G_{\mathbf{k}}^{-1}(i\omega) = i\omega - \epsilon(\mathbf{k}), \quad (\text{S40})$$

$$V[A] = e\mathbf{A} \cdot \mathbf{v}(\mathbf{k}) + e\mathbf{E} \cdot \mathbf{a}(\mathbf{k}) = e\mathbf{A} \cdot \mathbf{v}(\mathbf{k}) + i\omega_n \mathbf{A}(i\omega_n) \cdot \mathbf{a}(\mathbf{k}). \quad (\text{S41})$$

The polarization tensor is defined at quadratic order in $\mathbf{A}$ in the expansion of the effective action $S_{\text{eff}}[A]$,

$$\Pi^{ij}(i\omega) = e^2 T \sum_{\nu} \int \frac{d^d \mathbf{k}}{(2\pi)^d} \text{Tr} [G_{\mathbf{k}}(i\nu) \gamma_i(k) G_{\mathbf{k}}(i\nu + i\omega) \gamma_j(k)]. \quad (\text{S42})$$

Here $\gamma_i(k) = \gamma_i^{(\text{v})}(k) + \gamma_i^{(\text{B})}(k)$ with $\gamma_i^{(\text{v})}(k) = v_i(\mathbf{k})$ and $\gamma_i^{(\text{B})}(k) = i\omega_n a_i(\mathbf{k})$. The conductivity follows from the antisymmetric part of the polarization tensor, $\sigma_{ij} = \lim_{\omega \rightarrow 0} \Pi^{ij,a}(\omega + i\eta, \mathbf{0})/(i\omega)$ with $\Pi^{ij,a} = \frac{1}{2}(\Pi^{ij} - \Pi^{ji})$. For the Hall conductivity, we calculate

$$\begin{aligned} \Pi_{ij}^{(\text{hall})}(i\omega_n, \mathbf{0}) &= e^2 T \sum_{\nu} \int \frac{d^d \mathbf{k}}{(2\pi)^d} \text{Tr} [G_{\mathbf{k}}(i\nu) \gamma_i^{(\text{v})} G_{\mathbf{k}}(i\nu + i\omega_n) \gamma_j^{(\text{B})}] - (i \leftrightarrow j) \\ &= i\omega_n e^2 T \sum_{\nu} \int \frac{d^d \mathbf{k}}{(2\pi)^d} [v_i(\mathbf{k}) a_j(\mathbf{k}) - v_j(\mathbf{k}) a_i(\mathbf{k})] G_{\mathbf{k}}(i\nu) G_{\mathbf{k}}(i\nu + i\omega_n) \\ &= i\omega_n e^2 \int \frac{d^d \mathbf{k}}{(2\pi)^d} [v_i(\mathbf{k}) a_j(\mathbf{k}) - v_j(\mathbf{k}) a_i(\mathbf{k})] \left[T \sum_{\nu} G_{\mathbf{k}}^2(i\nu) \right] + \mathcal{O}(\omega_n^2) \\ &= -i\omega_n e^2 \int \frac{d^d \mathbf{k}}{(2\pi)^d} [v_i(\mathbf{k}) a_j(\mathbf{k}) - v_j(\mathbf{k}) a_i(\mathbf{k})] \frac{\partial n_F(\epsilon_{\mathbf{k}})}{\partial \epsilon_{\mathbf{k}}} + \mathcal{O}(\omega_n^2), \end{aligned} \quad (\text{S43})$$

where we expand $G_{\mathbf{k}}(i\nu + i\omega_n)$ at small ω_n : $G_{\mathbf{k}}(i\nu + i\omega_n) = G_{\mathbf{k}}(i\nu) - i\omega_n G_{\mathbf{k}}^2(i\nu) + \mathcal{O}(\omega_n^2)$ and use $T \sum_{\nu} G_{\mathbf{k}}^2(i\nu) = -\frac{\partial n_F(\epsilon_{\mathbf{k}})}{\partial \epsilon_{\mathbf{k}}}$. Noting that

$$\begin{aligned} &\int \frac{d^d \mathbf{k}}{(2\pi)^d} [v_i(\mathbf{k}) a_j(\mathbf{k}) - v_j(\mathbf{k}) a_i(\mathbf{k})] \frac{\partial n_F(\epsilon_{\mathbf{k}})}{\partial \epsilon_{\mathbf{k}}} \\ &= \int \frac{d^d \mathbf{k}}{(2\pi)^d} \left[\frac{\partial \epsilon_{\mathbf{k}}}{\partial k_i} a_j(\mathbf{k}) - \frac{\partial \epsilon_{\mathbf{k}}}{\partial k_j} a_i(\mathbf{k}) \right] \frac{\partial n_F(\epsilon_{\mathbf{k}})}{\partial \epsilon_{\mathbf{k}}} \end{aligned} \quad (\text{S44})$$

$$= - \int \frac{d^d \mathbf{k}}{(2\pi)^d} n_F(\epsilon_{\mathbf{k}}) \Omega_{ij}(\mathbf{k}), \quad (\text{S45})$$

we obtain the Hall conductivity

$$\sigma_{xy} = \lim_{\omega \rightarrow 0} \frac{1}{i\omega} \Pi^{xy,a}(\omega) = \frac{e^2}{h} \int \frac{d^d \mathbf{k}}{(2\pi)^d} n_F(\epsilon_{\mathbf{k}}) \Omega_{xy}(\mathbf{k}). \quad (\text{S46})$$

S3. NOZIÈRES-LUTTINGER CONSTRUCTION

In this section, we apply the Nozières-Luttinger construction to the BL-FL theory.

We first review the construction, which identifies the low-energy processes responsible for the collective response of a Fermi liquid. Particle-hole pairs near the Fermi surface carrying a small momentum $\mathbf{q}$ and frequency ω give rise to the singular dependence on the frequency ω and momentum $\mathbf{q}$. In contrast, the regular interaction processes are collected into an effective scattering vertex I . Diagrammatically, I is irreducible in the relevant particle-hole channel: it cannot be split into two pieces by cutting the propagators of the particle-hole pair. The full vertex Γ satisfies the Bethe-Salpeter equation,

$$\Gamma(\mathbf{k}, \mathbf{k}'; i\omega, \mathbf{q}) = I_{\mathbf{k}\mathbf{k}'} + \int_{\mathbf{p}} I_{\mathbf{k}\mathbf{p}} R(\mathbf{p}; i\omega, \mathbf{q}) \Gamma(\mathbf{p}, \mathbf{k}'; i\omega, \mathbf{q}), \quad (\text{S47})$$

where R is the bare particle-hole kernel

$$R(\mathbf{p}; i\omega, \mathbf{q}) = T \sum_{\nu} G(\mathbf{p} + \mathbf{q}, i\nu + i\omega) G(\mathbf{p}, i\nu) = \frac{n_{\mathbf{p}} - n_{\mathbf{p}+\mathbf{q}}}{i\omega + \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{p}+\mathbf{q}}}, \quad (\text{S48})$$

with $n_{\mathbf{p}} = n_F(\epsilon_{\mathbf{p}})$. For instance, given the density-density interaction in Eq. (S28), the bare particle-hole irreducible vertex is $I_{\mathbf{k}\mathbf{k}'}^0 = -V_{\mathbf{k}-\mathbf{k}'} |\langle u_{\mathbf{k}} | u_{\mathbf{k}'} \rangle|^2$, while I is obtained by including the interaction renormalization of the bare vertex I^0 .

In operator notation, the Bethe-Salpeter equation is $\Gamma = I + IR\Gamma$, or equivalently $\Gamma = (1 - IR)^{-1}I$. Near $(\omega, \mathbf{q}) = (0, \mathbf{0})$, the vertex Γ depends on the order in which two limits are taken: the dynamic limit

$$\Gamma^{\omega}(\mathbf{k}, \mathbf{k}') = \lim_{\omega \rightarrow 0} \lim_{\mathbf{q} \rightarrow 0} \Gamma(\mathbf{k}, \mathbf{k}'; i\omega, \mathbf{q}),$$

and the static limit

$$\Gamma^q(\mathbf{k}, \mathbf{k}') = \lim_{\mathbf{q} \rightarrow 0} \lim_{\omega \rightarrow 0} \Gamma(\mathbf{k}, \mathbf{k}'; i\omega, \mathbf{q}).$$

For the kernel defined in Eq. (S48), the dynamic limit is $R^{\omega}(\mathbf{p}) = \lim_{\omega \rightarrow 0} \lim_{\mathbf{q} \rightarrow 0} R(\mathbf{p}; i\omega, \mathbf{q}) = 0$. Thus, we have $\Gamma_{\mathbf{k}\mathbf{k}'}^{\omega} = I_{\mathbf{k}\mathbf{k}'}$.

The Landau interaction is defined from the vertex $\Gamma_{\mathbf{k}\mathbf{k}'}^{\omega}$ in the dynamic limit

$$f_{\mathbf{k}\mathbf{k}'} = Z_{\mathbf{k}} Z_{\mathbf{k}'} \Gamma_{\mathbf{k}\mathbf{k}'}^{\omega}, \quad (\text{S49})$$

where $Z_{\mathbf{k}}$ is the quasiparticle weight. In the static limit, the particle-hole kernel becomes

$$R^q(\mathbf{p}) = \lim_{\mathbf{q} \rightarrow 0} \lim_{\omega \rightarrow 0} R(\mathbf{p}; i\omega, \mathbf{q}) = \partial_{\epsilon} n_F(\epsilon_{\mathbf{p}}) \equiv -\chi_{\mathbf{p}}.$$

The Bethe-Salpeter equation becomes $\Gamma^q = I - I\chi\Gamma^q$. Introducing the renormalized static vertex $\tilde{\Gamma}_{\mathbf{k}\mathbf{k}'}^q = Z_{\mathbf{k}} Z_{\mathbf{k}'} \Gamma_{\mathbf{k}\mathbf{k}'}^q$, and using Eq. (S49), we obtain

$$\tilde{\Gamma}_{\mathbf{k}\mathbf{k}'}^q = f_{\mathbf{k}\mathbf{k}'} - \int_{\mathbf{p}} f_{\mathbf{k}\mathbf{p}} \chi_{\mathbf{p}} \tilde{\Gamma}_{\mathbf{p}\mathbf{k}'}^q.$$

Thus $f_{\mathbf{k}\mathbf{k}'}$ is the interaction between quasiparticles in the dynamic limit, while the static vertex contains additional dressing from quasiparticles near the Fermi surface.

A. Bethe-Salpeter equations for the electromagnetic vertex

A subtlety arises because the interaction generates not only the particle-hole ladder but also an interaction-induced contribution to the current operator. Thus, the quantum-geometric current forms part of the irreducible electromagnetic vertex.

Starting from the full Green function in an external vector potential, $G^{-1}[A] = G_0^{-1}[A] - \Sigma[G, A]$, the electromagnetic vertex is obtained by differentiating with respect to $\mathbf{A}$,

$$\frac{\delta G^{-1}[A]}{\delta A_i} = \frac{\delta G_0^{-1}[A]}{\delta A_i} - \left(\frac{\delta \Sigma[G, A]}{\delta A_i} \right)_G - \left(\frac{\delta \Sigma[G, A]}{\delta G} \right)_A \frac{\delta G}{\delta A_i}. \quad (\text{S50})$$

The anomalous potential term will be discussed later. Defining the full electromagnetic vertex $\Lambda_i = -\frac{\delta G^{-1}[A]}{\delta A_i}$, and using $\frac{\delta G}{\delta A_i} = G\Lambda_i G$, we have the Bethe-Salpeter equation

$$\Lambda_i = \gamma_i^{(\text{v})} + \gamma_i^{(\text{int})} + I R \Lambda_i.$$

Here $\gamma_i^{(\text{v})}(\mathbf{k}) = -\frac{\delta G_0^{-1}[A]}{\delta A_i} = e\partial_{k_i}\epsilon_{\mathbf{k}}^0$ is the bare velocity vertex, $\gamma_i^{(\text{int})} = \left(\frac{\delta \Sigma[G, A]}{\delta A_i} \right)_G$ is the interaction-induced irreducible quantum-geometric current vertex, $I = \left(\frac{\delta \Sigma[G, A]}{\delta G} \right)_A$ is the particle-hole irreducible four-point vertex, and R is the bare particle-hole kernel.

We first solve the Bethe-Salpeter equation in the dynamic limit. Since $R^\omega = 0$, we have $\Lambda_i = \gamma_i^{(\text{v})} + \gamma_i^{(\text{int})}$. With $f_{\mathbf{k}\mathbf{k}'} = Z_{\mathbf{k}}Z_{\mathbf{k}'}\Gamma_{\mathbf{k}\mathbf{k}'}^\omega$, the interaction-induced quantum-geometric current vertex becomes $\gamma_i^{(\text{int}, \omega)} = e \int_{\mathbf{k}'} n_{\mathbf{k}'} (\partial_{k_i} + \partial_{k'_i}) f_{\mathbf{k}\mathbf{k}'}$. Therefore, we obtain the physical current dressed by Landau backflow,

$$\mathcal{J}_i^\omega = ev_i^0 + \gamma_i^{(\text{int}, \omega)} = (1 + f\chi)ev_i^{\text{qp}}$$

with v_i^{qp} being the quasiparticle velocity. In the static limit, $R^q = -\chi$. The Bethe-Salpeter equation becomes $\mathcal{J}_i^q = \mathcal{J}_i^\omega - f\chi\mathcal{J}_i^q$, giving $\mathcal{J}_i^q = ev_i^{\text{qp}}$.

We then discuss the anomalous potential term with the bare vertex $\gamma_i^{(\text{B})}(\mathbf{k}) = i\omega a_i(\mathbf{k})$. We factor out the explicit $i\omega$ by defining $\Lambda_i^{(\text{B})} = i\omega \mathcal{A}_i(\mathbf{k})$ with the full vertex $\Lambda_i^{(\text{B})}$ dressed by interactions. The Bethe-Salpeter equation reduces to

$$\mathcal{A}_i(\mathbf{k}; i\omega, \mathbf{q}) = a_i(\mathbf{k}) + \int_{\mathbf{p}} I_{\mathbf{k}\mathbf{p}} R(\mathbf{p}; i\omega, \mathbf{q}) \mathcal{A}_i(\mathbf{p}; i\omega, \mathbf{q}). \quad (\text{S51})$$

In the static limit $R^q = -\chi$, we have $\mathcal{A}_i^q = (1 + f\chi)^{-1}a_i$. In the dynamic limit, we have $\mathcal{A}_i^\omega = a_i$.

S4. AHE AND DRUDE WEIGHT IN THE BL-FL THEORY

In this section, we derive the Drude weight and intrinsic anomalous Hall response from the energy functional of the BL-FL theory.

A. Response theory from the free energy

We start with the free energy

$$F[n, \mathbf{A}] = \delta E_{\text{BL-FL}}[n, \mathbf{A}] - TS[n],$$

with the entropy

$$S[n] = - \sum_{\mathbf{k}} [n_{\mathbf{k}} \ln n_{\mathbf{k}} + (1 - n_{\mathbf{k}}) \ln(1 - n_{\mathbf{k}})].$$

We expand the distribution function around equilibrium $n_{\mathbf{k}} = n_{\mathbf{k}}^0 + \delta n_{\mathbf{k}}$ and determine the equilibrium distribution $n_{\mathbf{k}}^0$ by minimizing the free energy $F[n, \mathbf{A}]$. This gives $n_{\mathbf{k}}^0 = n_F(\epsilon_{\mathbf{k}}^{\text{qp}} - \mu)$. The fluctuation part of the free energy $\delta F = F[n, \mathbf{A}] - F[n^0, 0]$ is given by

$$\begin{aligned} \delta F = & \frac{1}{2} \int_{\mathbf{k}} \chi_{\mathbf{k}}^{-1} (\delta n_{\mathbf{k}})^2 + \frac{1}{2} \int_{\mathbf{k}\mathbf{k}'} f_{\mathbf{k}\mathbf{k}'} \delta n_{\mathbf{k}} \delta n_{\mathbf{k}'} \\ & + \int_{\mathbf{k}} [\mathbf{A} \cdot \mathcal{J}^\omega(\mathbf{k}) + e\mathbf{E} \cdot \mathbf{a}(\mathbf{k})] \delta n_{\mathbf{k}} + \frac{1}{2} \int_{\mathbf{k}\mathbf{k}'} \mathbf{A} \cdot \mathbf{\Lambda}_{\mathbf{k}\mathbf{k}'} \delta n_{\mathbf{k}} \delta n_{\mathbf{k}'}. \end{aligned} \quad (\text{S52})$$

Here $\mathbf{a}(\mathbf{k})$ is the Berry connection of the quasiparticles and $\mathcal{J}^\omega(\mathbf{k})$ is the dynamic-limit current,

$$\mathcal{J}^\omega(\mathbf{k}) = e\mathbf{v}^0(\mathbf{k}) + e \int_{\mathbf{k}'} (\nabla_{\mathbf{k}} + \nabla_{\mathbf{k}'}) f_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}'}^0.$$

The term $\frac{1}{2} \int_{\mathbf{k}} \chi_{\mathbf{k}}^{-1} (\delta n_{\mathbf{k}})^2$ with $\chi_{\mathbf{k}} = \beta n_{\mathbf{k}}^0 (1 - n_{\mathbf{k}}^0)$ arises from the entropy term. We minimize δF with respect to $\delta n_{\mathbf{k}}$. To linear order in the external fields, we obtain the equation

$$\chi_{\mathbf{k}}^{-1} \delta n_{\mathbf{k}} + \int_{\mathbf{k}'} f_{\mathbf{k}\mathbf{k}'} \delta n_{\mathbf{k}'} = -\mathbf{A} \cdot \mathcal{J}^\omega(\mathbf{k}) - e\mathbf{E} \cdot \mathbf{a}(\mathbf{k}). \quad (\text{S53})$$

The term $\mathbf{A} \cdot \mathbf{\Lambda}_{\mathbf{k}\mathbf{k}'}$ is dropped at leading order. If $M_{\mathbf{k}\mathbf{k}'}$ denotes the inverse of the quadratic kernel

$$\int_{\mathbf{p}} [\chi_{\mathbf{k}}^{-1} \delta_{\mathbf{k}\mathbf{p}} + f_{\mathbf{k}\mathbf{p}}] M_{\mathbf{p}\mathbf{k}'} = \delta_{\mathbf{k}\mathbf{k}'}, \quad (\text{S54})$$

we get the expression for $\delta n_{\mathbf{k}}$ in terms of the gauge field

$$\delta n_{\mathbf{k}} = - \int_{\mathbf{k}'} M_{\mathbf{k}\mathbf{k}'} [\mathbf{A} \cdot \mathcal{J}^\omega(\mathbf{k}') + e\mathbf{E} \cdot \mathbf{a}(\mathbf{k}')]. \quad (\text{S55})$$

Thus, we have the free energy to quadratic order

$$\delta F = -\frac{1}{2} \int_{\mathbf{k}\mathbf{k}'} s_{\mathbf{k}} M_{\mathbf{k}\mathbf{k}'} s_{\mathbf{k}'} + \mathcal{O}(A^3, A^2 E, A E^2), \quad (\text{S56})$$

where $s_{\mathbf{k}} = \mathbf{A} \cdot \mathcal{J}^\omega(\mathbf{k}) + e\mathbf{E} \cdot \mathbf{a}(\mathbf{k})$. The $\mathbf{A} \cdot \mathbf{\Lambda}_{\mathbf{k}\mathbf{k}'}$ term in Eq. (S52) does not modify the Drude weight or the linear AHE response. It can contribute to the nonlinear response.

Drude weight.—The $A_i A_j$ term of Eq. (S56) determines the Drude weight. In matrix form, the response kernel is

$$M = (\chi^{-1} + f)^{-1} = \chi(1 + f\chi)^{-1}$$

from Eq. (S54). The Drude weight can be obtained from the derivatives of the free energy $D_{ij} = \frac{\delta^2 F}{\delta A_i \delta A_j}$, giving

$$D_{ij} = e^2 \int_{\mathbf{k}} \chi_{\mathbf{k}} v_{\mathbf{k},i}^{\text{qp}} v_{\mathbf{k},j}^{\text{qp}} + e^2 \int_{\mathbf{k}\mathbf{k}'} \chi_{\mathbf{k}} v_i^{\text{qp}}(\mathbf{k}) \chi_{\mathbf{k}'} f_{\mathbf{k}\mathbf{k}'} v_{\mathbf{k}',j}^{\text{qp}}. \quad (\text{S57})$$

For an isotropic Fermi liquid, $D(0) = D^{\text{qp}} (1 + \frac{F_1}{d})$, with F_1 being the $\ell = 1$ Landau parameter. Here D^{qp} is the Drude weight associated with the quasiparticles without the Landau backflow.

Anomalous Hall response.—The mixed $A_i E_j$ term of Eq. (S56) determines the anomalous Hall conductivity $\sigma_{ij}^{\text{AHE}} = \frac{\delta^2 F}{\delta A_i \delta E_j} - \frac{\delta^2 F}{\delta A_j \delta E_i}$. Explicitly, we have

$$\sigma_{ij}^{\text{AHE}} = e \left[\int_{\mathbf{k}\mathbf{k}'} \mathcal{J}_i^\omega(\mathbf{k}) M_{\mathbf{k}\mathbf{k}'} a_j(\mathbf{k}') - (i \leftrightarrow j) \right].$$

The Landau-interaction dressing cancels completely between $\mathcal{J}^\omega = (1 + f\chi)e\mathbf{v}^{\text{qp}}$ and $M = \chi(1 + f\chi)^{-1}$. Therefore,

$$\begin{aligned} \sigma_{ij}^{\text{AHE}} &= e^2 \int_{\mathbf{k}} \chi_{\mathbf{k}} \left[v_{\mathbf{k},i}^{\text{qp}} a_j(\mathbf{k}) - v_{\mathbf{k},j}^{\text{qp}} a_i(\mathbf{k}) \right] \\ &= \frac{e^2}{\hbar} \int_{\mathbf{k}} n_F(\epsilon_{\mathbf{k}}^{\text{qp}} - \mu) \Omega_{ij}^{\text{qp}}(\mathbf{k}). \end{aligned} \quad (\text{S58})$$

Here $\hbar$ has been restored in the second line. Thus, the anomalous Hall conductivity contains no additional Landau backflow factor. Interactions can still affect the Hall response indirectly through the renormalized quasiparticle dispersion and Berry curvature. Including the fully occupied remote bands, $\sigma_{ij}^{\text{AHE}} = C_{\text{filled}} \frac{e^2}{\hbar} + \frac{e^2}{\hbar} \int_{\text{active}} n_F(\epsilon_{\mathbf{k}}^{\text{qp}} - \mu) \Omega_{ij}^{\text{qp}}(\mathbf{k})$.

S5. GENERIC PERTURBATIONS

A. General formula

In this section, we discuss a generalized Peierls substitution for a generic response. Consider a slow external force $X_\alpha(t)$ that deforms the quasiparticle momentum as

$$\mathbf{k}^X = \mathbf{k} + \sum_{\alpha} \mathbf{G}_{\alpha}(\mathbf{k}) X_{\alpha},$$

where α labels the components of the external source and $\mathbf{G}_{\alpha}(\mathbf{k})$ specifies how the source moves a state at momentum $\mathbf{k}$. For a spatially uniform, slowly varying source, the instantaneous Bloch state follows the corresponding generalized Houston trajectory, $|u_{\mathbf{k}}(t)\rangle = |u_{\mathbf{k}(t)}\rangle$, where $\mathbf{k}(t)$ satisfies the equation of motion $\dot{\mathbf{k}} = \sum_{\alpha} \mathbf{G}_{\alpha}(\mathbf{k}) \dot{X}_{\alpha}$. Then we obtain the anomalous potential term

$$L_B = \sum_{\mathbf{k}} \dot{\mathbf{k}} \cdot \mathbf{a}(\mathbf{k}) f_{\mathbf{k}}^{\dagger} f_{\mathbf{k}} = \sum_{\mathbf{k}} \sum_{\alpha} \dot{X}_{\alpha} \mathbf{G}_{\alpha}(\mathbf{k}) \cdot \mathbf{a}(\mathbf{k}) f_{\mathbf{k}}^{\dagger} f_{\mathbf{k}}. \quad (\text{S59})$$

On the other hand, perturbation theory gives the change of the Bloch wave under the external force,

$$|\delta u_{\mathbf{k}}\rangle = \sum_{i\alpha} X_{\alpha} G_{i\alpha}(\mathbf{k}) |\mathcal{D}_i u_{\mathbf{k}}\rangle,$$

where $\mathcal{D}_i = \partial_{k_i} - ia_i(\mathbf{k})$ is the covariant derivative. Given the projected density-density interaction in the forward-scattering channel $H_{\text{int}} \rightarrow f_{\mathbf{k}\mathbf{k}'}^0 n_{\mathbf{k}} n_{\mathbf{k}'}$, with $f_{\mathbf{k}\mathbf{k}'}^0 = -V_{\mathbf{k}-\mathbf{k}'} |\langle u_{\mathbf{k}} | u_{\mathbf{k}'} \rangle|^2$, we have the interaction-induced current operator,

$$J_{\alpha}^X(\mathbf{k}) = -\frac{1}{V} \sum_{\mathbf{k}'} n_{\mathbf{k}} n_{\mathbf{k}'} \sum_i [G_{i\alpha}(\mathbf{k}) \partial_{k_i} + G_{i\alpha}(\mathbf{k}') \partial_{k'_i}] f_{\mathbf{k}\mathbf{k}'}^0. \quad (\text{S60})$$

Within the BL-FL theory, this current contributes the following term to the energy functional

$$\delta E[n, X] = \frac{1}{V} \sum_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}} n_{\mathbf{k}'} \sum_{i\alpha} X_{\alpha} [G_{i\alpha}(\mathbf{k}) \partial_{k_i} + G_{i\alpha}(\mathbf{k}') \partial_{k'_i}] f_{\mathbf{k}\mathbf{k}'}. \quad (\text{S61})$$

Consider a vector potential $\mathbf{A}^{\lambda}$ that shifts momentum according to $\mathbf{k} \rightarrow \mathbf{k} + \lambda_{\mathbf{k}} \mathbf{A}^{\lambda}$, that is, $G_{i\alpha}(\mathbf{k}) = \lambda_{\mathbf{k}} \delta_{\alpha i}$. The current operator in Eq. (S60) becomes

$$J_i^{\lambda}(\mathbf{k}) = -\frac{1}{V} \sum_{\mathbf{k}'} n_{\mathbf{k}} n_{\mathbf{k}'} [\lambda_{\mathbf{k}} \partial_{k_i} + \lambda_{\mathbf{k}'} \partial_{k'_i}] f_{\mathbf{k}\mathbf{k}'}^0, \quad (\text{S62})$$

and the corresponding term in the BL-FL energy functional is

$$\delta E[n, \lambda] = \frac{1}{V} \sum_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}} n_{\mathbf{k}'} \mathbf{A}^{\lambda} \cdot [\lambda_{\mathbf{k}} \nabla_{\mathbf{k}} + \lambda_{\mathbf{k}'} \nabla_{\mathbf{k}'}] f_{\mathbf{k}\mathbf{k}'}. \quad (\text{S63})$$

1. Thermal current

For the thermal current, Luttinger's gravitational vector potential $\mathbf{A}^g$ produces the low-energy shift $\mathbf{k} \rightarrow \mathbf{k} + \xi_{\mathbf{k}} \mathbf{A}^g$ with $\xi_{\mathbf{k}} = \epsilon_{\mathbf{k}}^{\text{qp}} - \mu$ and $\lambda_{\mathbf{k}} = \xi_{\mathbf{k}}$. In the BL-FL theory, the interaction-induced heat current is

$$\mathcal{J}_{\alpha}^{Q, \text{int}}(\mathbf{k}) = \frac{1}{V} \sum_{\mathbf{k}'} n_{\mathbf{k}'} [\xi_{\mathbf{k}} \partial_{k_{\alpha}} + \xi_{\mathbf{k}'} \partial_{k'_{\alpha}}] f_{\mathbf{k}\mathbf{k}'} \quad (\text{S64})$$

$$= \frac{1}{V} \sum_{\mathbf{k}'} n_{\mathbf{k}'} \left[\frac{\xi_{\mathbf{k}} + \xi_{\mathbf{k}'}}{2} (\partial_{k_{\alpha}} + \partial_{k'_{\alpha}}) + \frac{\xi_{\mathbf{k}} - \xi_{\mathbf{k}'}}{2} (\partial_{k_{\alpha}} - \partial_{k'_{\alpha}}) \right] f_{\mathbf{k}\mathbf{k}'}. \quad (\text{S65})$$

For a slowly varying gravitational vector potential, $\mathbf{E}^g = -\partial_t \mathbf{A}^g = -\frac{\nabla T}{T}$, and the crystal momentum $\mathbf{k}$ follows the equation of motion $\dot{\mathbf{k}} = -\xi_{\mathbf{k}} \frac{\nabla T}{T}$. The corresponding anomalous potential is

$$\delta L_B = \frac{\xi_{\mathbf{k}}}{T} \nabla T \cdot \mathbf{a}(\mathbf{k}).$$

This gives the thermal analogue of the Wannier-Stark energy contributed by the Berry connection.

2. Frame rotation

A rotation of the local frame by an angle $\theta(t)$ about $\hat{z}$ acts on momentum as

$$\mathbf{k}_\theta = R_\theta^{-1} \mathbf{k} = \mathbf{k} - \theta \hat{z} \times \mathbf{k} + \mathcal{O}(\theta^2). \quad (\text{S66})$$

Thus, $\dot{\mathbf{k}} = -\omega_z \hat{z} \times \mathbf{k}$ with $\omega_z = \dot{\theta}$. If the projected basis also transforms under an internal rotation $|u_{\mathbf{k}}; \theta\rangle = e^{-i\theta S_z^{\text{orbital}}} |u_{R_\theta^{-1}\mathbf{k}}\rangle$, the spatial rotation generates an additional internal spin-connection term. Thus, we have the anomalous potential term

$$\delta L_B = -\omega_z (\hat{z} \times \mathbf{k}) \cdot \mathbf{a}(\mathbf{k}) - \omega_z s_z(\mathbf{k}) \quad (\text{S67})$$

with the projected spin connection $s_z(\mathbf{k}) = \langle u_{\mathbf{k}} | S_z^{\text{orbital}} | u_{\mathbf{k}} \rangle$. In the interaction, the internal unitary rotation cancels in the form factor $\langle u_{\mathbf{k}}; \theta | u_{\mathbf{k}}; \theta \rangle = \langle u_{R_\theta^{-1}\mathbf{k}} | u_{R_\theta^{-1}\mathbf{k}} \rangle$. Thus, the interaction-induced rotation current operator is

$$J_z^\theta(\mathbf{k}) = -\frac{1}{V} \sum_{\mathbf{k}'} n_{\mathbf{k}} n_{\mathbf{k}'} [(\hat{z} \times \mathbf{k}) \cdot \nabla_{\mathbf{k}} + (\hat{z} \times \mathbf{k}') \cdot \nabla_{\mathbf{k}'}] f_{\mathbf{k}\mathbf{k}'}^0. \quad (\text{S68})$$

3. Vielbein

We start with the perturbation from the vielbein field to consider the forces due to strain, shear, and rotation. We take a time-dependent spatial vielbein $e_a^i(t)$ and its inverse $e_a^i(t)$: $e_a^i e_i^b = \delta_a^b$. Let q_i be the conserved coordinate momentum; then the momentum entering the local Bloch Hamiltonian is $k_a(t) = e_a^i q_i$. From $\dot{k}_a = \dot{e}_a^i q_i$ and $\dot{e}_a^i = -e_a^j \dot{e}_j^b e_b^i$, we obtain

$$\dot{k}_a = -\Gamma_{ab} k_b, \quad \Gamma_{ab} \equiv e_a^i \dot{e}_i^b. \quad (\text{S69})$$

For a weak deformation of the undeformed frame, $e_a^i \simeq \delta_a^i$, and we simply have $\dot{k}_i = -\Gamma_{ij} k_j$. Hence, we have the vielbein-induced Berry term

$$\delta L_B = -\sum_{\mathbf{k}} \Gamma_{ij} k_j a_i(\mathbf{k}) f_{\mathbf{k}}^\dagger f_{\mathbf{k}}. \quad (\text{S70})$$

We can decompose Γ_{ij} into symmetric and antisymmetric parts: $\Gamma_{ij} = \Gamma_{(ij)} + \Gamma_{[ij]}$ with $\Gamma_{(ij)} = \frac{1}{2}(\Gamma_{ij} + \Gamma_{ji})$ and $\Gamma_{[ij]} = \frac{1}{2}(\Gamma_{ij} - \Gamma_{ji})$. The symmetric part describes the strain and shear u_{ij} with the relation $\Gamma_{(ij)} = \dot{u}_{ij}$, giving $\dot{k}_i = -\dot{u}_{ij} k_j$. The corresponding term in BL-FL is

$$\delta E[n, u] = \frac{1}{V} \sum_{\mathbf{k}\mathbf{k}'} n_{\mathbf{k}} n_{\mathbf{k}'} \sum_{ij} u_{ij} (k_j \partial_{k_i} + k_i \partial_{k_j}) f_{\mathbf{k}\mathbf{k}'}. \quad (\text{S71})$$

Rotation is described by the antisymmetric part $\Gamma_{[ij]} = \omega_z \epsilon_{ij}$.

S6. NUMERICS ON THE WILSON-DIRAC MODEL

We give the numerical details of the Wilson-Dirac model and the self-consistent Hartree-Fock calculation used to test the BL-FL theory.

A. Wilson-Dirac model

We consider spinless fermions with two orbitals per unit cell, $H_0 = \sum_{\mathbf{k}} c_{\mathbf{k}}^\dagger H_0(\mathbf{k}) c_{\mathbf{k}}$, $c_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}1} \\ c_{\mathbf{k}2} \end{pmatrix}$. The Bloch projectors are obtained from the vector $\mathbf{d}(\mathbf{k})$ with components

$$d_x(\mathbf{k}) = v \sin k_x, \quad d_y(\mathbf{k}) = v \sin k_y, \quad d_z(\mathbf{k}) = M - 2B(2 - \cos k_x - \cos k_y), \quad (\text{S72})$$

and $\hat{\mathbf{d}}(\mathbf{k}) = \frac{\mathbf{d}(\mathbf{k})}{|\mathbf{d}(\mathbf{k})|}$. For $0 < M < 4B$, the two bands carry opposite nonzero Chern numbers. To control the band dispersion and the remote-band gap independently, we use a deformed Wilson-Dirac Hamiltonian

$$H_0(\mathbf{k}) = s\zeta(\mathbf{k})\sigma_0 + \Delta\hat{\mathbf{d}}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad (\text{S73})$$

with $\zeta(\mathbf{k}) = |\mathbf{d}(\mathbf{k})|$. The eigenvalues of H_0 are $\epsilon_{\pm}^0(\mathbf{k}) = s\zeta(\mathbf{k}) \pm \Delta$. Here s controls the bandwidth, while 2Δ is the band gap. Thus, we can tune the band gap and bandwidth while keeping the quantum geometry fixed. We consider a screened density-density interaction

$$H_{\text{int}} = \frac{1}{2N} \sum_{\mathbf{q}} V(\mathbf{q}) \rho_{\mathbf{q}} \rho_{-\mathbf{q}}, \quad (\text{S74})$$

where $\rho_{\mathbf{q}} = \sum_{\mathbf{k}, \alpha} c_{\mathbf{k}+\mathbf{q}, \alpha}^\dagger c_{\mathbf{k}, \alpha}$ and $V(\mathbf{q}) = \frac{g}{1+2\ell_s^2(2-\cos q_x - \cos q_y)}$. We take the lower band to be fully filled and tune the chemical potential through the upper band. After projection onto the active band, the HF energy of the upper band, $\epsilon_{\mathbf{k}}^0 \equiv \epsilon_{+}^0(\mathbf{k})$, is

$$\epsilon_{\mathbf{k}}^{\text{HF}} = \epsilon_{\mathbf{k}}^0 + \frac{1}{N} \sum_{\mathbf{k}'} f_{\mathbf{k}\mathbf{k}'}^{\text{HF}} n_{\mathbf{k}'}^{\text{HF}}, \quad (\text{S75})$$

where $n_{\mathbf{k}}^{\text{HF}} = n_F(\epsilon_{\mathbf{k}}^{\text{HF}} - \mu)$. The Landau function in the HF approximation is

$$f_{\mathbf{k}\mathbf{k}'}^{\text{HF}} = V(0) - V(\mathbf{k} - \mathbf{k}') \text{Tr} [P_{\mathbf{k}}^0 P_{\mathbf{k}'}^0]. \quad (\text{S76})$$

The interaction contribution to the conserved HF current is

$$\gamma_i^{(\text{int}), \text{HF}}(\mathbf{k}) = \frac{e}{N} \sum_{\mathbf{k}'} n_{\mathbf{k}'}^{\text{HF}} (\partial_{k_i} + \partial_{k'_i}) f_{\mathbf{k}\mathbf{k}'}^{\text{HF}} \quad (\text{S77})$$

$$= -\frac{e}{N} \sum_{\mathbf{k}'} n_{\mathbf{k}'}^{\text{HF}} V(\mathbf{k} - \mathbf{k}') (\partial_{k_i} + \partial_{k'_i}) \text{Tr} [P_{\mathbf{k}}^0 P_{\mathbf{k}'}^0]. \quad (\text{S78})$$

Eqs. (S75) and (S76) are iterated until the change of the HF energy is below a fixed numerical tolerance.

B. Drude weight

The Drude weight is evaluated in two independent ways. First, we apply a uniform vector potential A_x . The bare dispersion and projected interaction kernel are both evaluated at the shifted Bloch momenta. The corresponding HF energy per unit cell is

$$E_{\text{HF}}(A_x) = \frac{1}{N} \sum_{\mathbf{k}} n_{\mathbf{k}}^{\text{HF}} \epsilon_{\mathbf{k}}^0(A_x) + \frac{1}{2N} \sum_{\mathbf{k}} n_{\mathbf{k}}^{\text{HF}} \Sigma_{\mathbf{k}}^{\text{HF}}(A_x), \quad (\text{S79})$$

with $\Sigma_{\mathbf{k}}^{\text{HF}}(A_x) = \frac{1}{N} \sum_{\mathbf{k}'} f_{\mathbf{k}\mathbf{k}'}^{\text{HF}}(A_x) n_{\mathbf{k}'}^{\text{HF}}$, and the Kohn stiffness is

$$D_{xx}^{\text{Kohn}} = \pi \left. \frac{\partial^2 F_{\text{HF}}(A_x)}{\partial A_x^2} \right|_{A_x=0}, \quad (\text{S80})$$

with the free energy $F_{\text{HF}}(A_x) = E_{\text{HF}}(A_x) - TS(A_x)$. Numerically, the curvature is extracted with a fourth-order symmetric five-point stencil. Second, we evaluate the finite-temperature Landau response. With $\chi_{\mathbf{k}} = -\frac{\partial n_F}{\partial \epsilon_{\mathbf{k}}^{\text{HF}}}$ and $v_{x,\mathbf{k}}^{\text{HF}} = \partial_{k_x} \epsilon_{\mathbf{k}}^{\text{HF}}$, the dressed velocity is

$$\mathcal{V}_{x,\mathbf{k}} = v_{x,\mathbf{k}}^{\text{HF}} + \frac{1}{N} \sum_{\mathbf{k}'} f_{\mathbf{k}\mathbf{k}'}^{\text{HF}} \chi_{\mathbf{k}'} v_{x,\mathbf{k}'}^{\text{HF}}.$$

The Drude weight is then

$$D_{xx} = \frac{\pi e^2}{N} \sum_{\mathbf{k}} \chi_{\mathbf{k}} v_{x,\mathbf{k}}^{\text{HF}} \mathcal{V}_{x,\mathbf{k}}. \quad (\text{S81})$$

Agreement between Eqs. (S80) and (S81) provides the numerical test shown in Fig. 1(c) of the main text.